

Job Aggregation Platform

Prepared for: Engineering & Product Management

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Dataset analyzed: jobs.json — 155,794 unique job records

Source documents: research/api_analysis.md, reports/analytics_summary.md

Table of Contents

Cover Page

2. Executive Summary

3. Project Overview

4. Dataset Overview

5. Provider Analysis

6. Regional Coverage

7. Job Freshness

8. Duplicate Analysis

9. API Analysis

10. India Research

11. Additional Regional APIs

12. API Filter Optimization

13. Maximum Coverage from Existing Providers

14. Recent Job Market Analysis (≤ 30 Days)

15. Engineering Recommendations

16. Regional API Expansion Strategy

17. Existing Provider Optimization

18. Regional Coverage Improvement Plan

19. ROI Prioritization

20. Roadmap

21. Future Improvements

22. Recent Job Market Analysis (Last 30 Days)

23. Conclusion

2. Executive Summary

The platform aggregates 155,794 unique job postings from 20 registered providers (19 contributing; TheirStack is blocked by an account-plan restriction) into one normalized schema, deduplicated by URL, and served through a static frontend.

Coverage is broad but concentrated: the US is 51% of all jobs, SmartRecruiters alone supplies 41.8% of the dataset, and one employer (Domino's, multi-location store postings) accounts for 15.7%.

Deduplication works correctly (zero duplicate URLs). Apparent same-title/company duplicates are legitimate multi-location roles, not errors — a naive rule would have erased tens of thousands of real listings.

Freshness is moderate but degrading: 54.0% of jobs were ≤ 30 days old at collection, 46.0% are stale, and 14.0% are over a year old, with no refresh/expiry mechanism to arrest the decline (see Section 7).

The highest-value opportunities identified are:

- Extend the existing, already-keyed Adzuna integration to additional countries - unlocking Canada, the UK, continental Europe, and Latin America with no new integration work.
- Introduce a scheduled refresh and expiry cycle to fix freshness.
- Close the Singapore/Southeast Asia and India coverage gaps via a no-auth government API and existing ATS infrastructure.
- Turn on date and location filters already supported by several integrated APIs but currently unused.

None of these require scraping, new commercial contracts, or code changes to the aggregation pipeline's core logic.

3. Project Overview

A modular Python pipeline (`jobs.py` + `sources/`) fetches every provider, normalizes records into a common seven-field schema, deduplicates by URL, and writes `jobs.json`. A static HTML/CSS/JS frontend reads that file directly.

Schema

Every record is normalized to seven fields — title, company, location, url, tags, remote, posted (ISO date) — letting 20 structurally different provider APIs feed one pipeline and frontend.

Provider architecture

Providers were added in three phases: Phase 1 (6 no-auth public APIs), Phase 2 (8 API-key providers, credentials in `.env`), Phase 3 (6 ATS integrations — Greenhouse, Lever, Ashby, SmartRecruiters, Workable, Teamtailor — one board per verified company). Every provider implements the same two-method interface (`fetch_raw`, `normalize`), so the core pipeline never changes.

Reliability

Each provider's `run()` method isolates failures — a bad response or normalization error is logged and skipped, not fatal. Retryable HTTP statuses (429/500/502/503/504) get exponential-backoff retries.

4. Dataset Overview

155,794 unique job records span 9,497 distinct companies. Schema completeness is very high: 0.03% missing a company name, 0.05% missing a location, 99.99% carry a parseable posting date.

Metric	Value
Total unique jobs	155,794
Registered providers	20
Providers contributing in latest run	19
Distinct companies	9,497
Jobs with empty company	45 (0.03%)
Jobs with empty location	72 (0.05%)
Jobs with empty tags	11,762 (7.5%)

Metric	Value
Jobs with a parseable posted date	155,779 (99.99%)
Duplicate URLs remaining	0

Remote vs. onsite

10.9% of jobs (16,992) are marked remote; 89.1% are onsite. Himalayas/RemoteOK/Jobicy are remote-only by construction; other providers use a genuine API field where available, else a keyword heuristic.

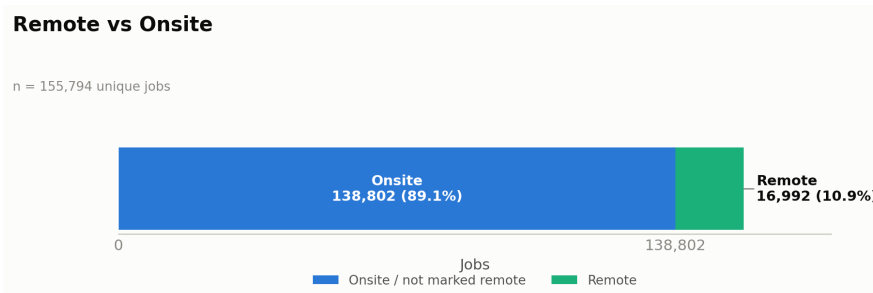


Figure 1. Remote vs. onsite split across the full dataset.

Company distribution

Company concentration is bimodal: the top 10 companies are 37.8% of the dataset (58,922 jobs), the top 25 are 47.5%, yet 58.0% of all 9,497 companies (5,505) contribute exactly one job. Domino's alone is 15.7% — its SmartRecruiters board lists each store-level opening separately across thousands of US locations.

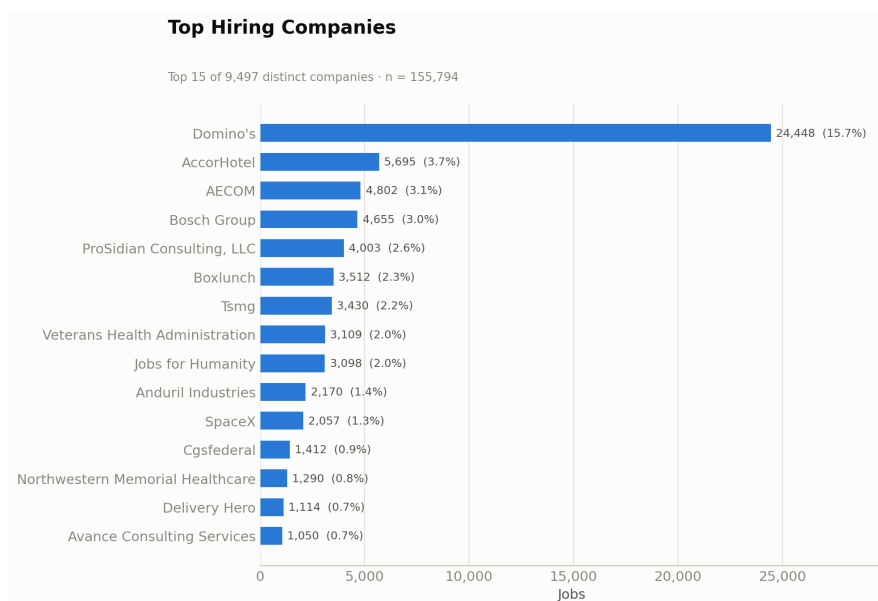


Figure 2. Top 15 hiring companies by job count.

5. Provider Analysis

Provider attribution is reconstructed from each job's URL host (the schema has no source field). SmartRecruiters alone supplies 41.8% of the dataset; the top three (SmartRecruiters, Lever, Greenhouse) supply 62.8%.

Provider	Jobs	% of dataset
SmartRecruiters	65,193	41.8%
Lever	17,755	11.4%
Greenhouse	14,930	9.6%
USAJOBS	10,000	6.4%
Reed	9,000	5.8%

Provider	Jobs	% of dataset
Bundesagentur (+AMS)	~7,800	~5.0%
Ashby	5,171	3.3%
Workable	4,159	2.7%
Jooble	~3,100	~2.0%
Himalayas	2,964	1.9%
Adzuna	2,001	1.3%
The Muse	2,000	1.3%
Arbeitsnow	930	0.6%
Teamtailor	474	0.3%
Other (SerpApi, OpenWeb Ninja, Careerjet, RemoteOK, Jobicy)	~350	~0.2%
TheirStack	0	0.0% (plan-gated, HTTP 402)

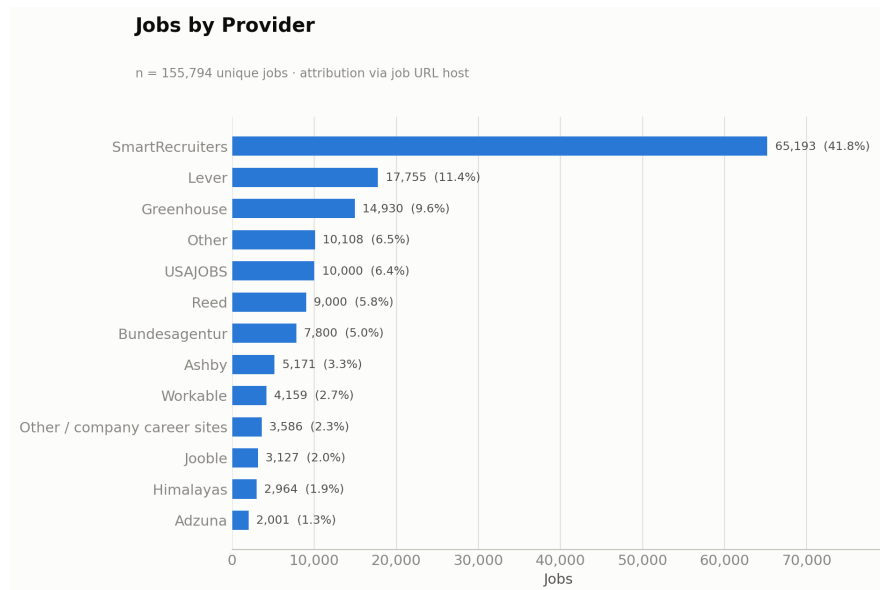


Figure 3. Jobs by provider (top 12, remainder grouped as Other).

ATS provider mix

The six ATS providers together supply 107,664 jobs (69.1% of the dataset): SmartRecruiters 60.6% of that subset, Lever 16.5%, Greenhouse 13.9%, and Ashby/Workable/Teamtailor the remaining 9.1%.

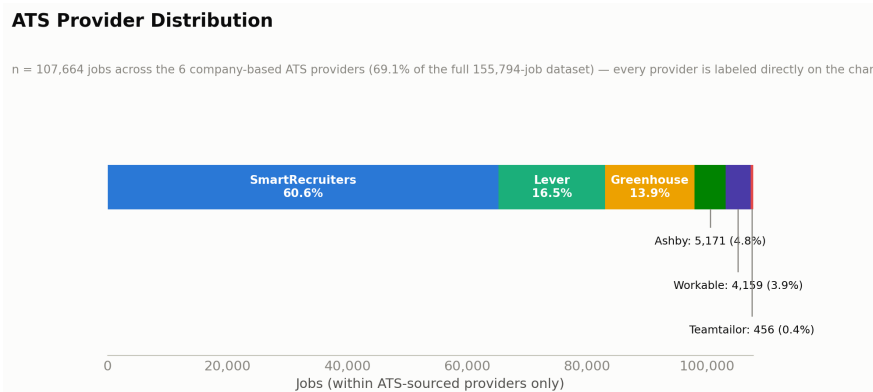


Figure 4. Distribution across the six ATS providers only.

All six ATS providers grow only by adding verified company slugs — there is no query-based expansion. Regional/company composition therefore reflects curation choices, not a balanced global crawl.

6. Regional Coverage

Region is inferred from free-text location via a keyword classifier. The US dominates at 51.0%; the six primary target regions together account for 81.3% (126,638 jobs).

Region	Jobs	% of dataset
United States	79,427	51.0%
Europe (continental, non-UK)	21,886	14.0%
United Kingdom	10,305	6.6%
LATAM	6,955	4.5%
Singapore / Southeast Asia	4,629	3.0%
Canada	3,436	2.2%
Remote / Worldwide	822	0.5%
Other / Unclassified	28,334	18.2%

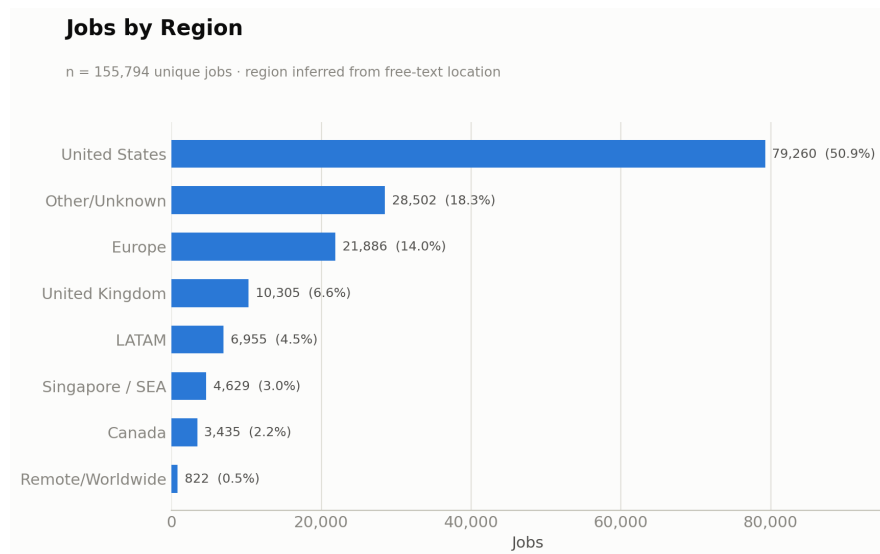


Figure 5. Jobs by region.

Canada is the weakest target region at 2.2% — under half the UK's coverage. 18.2% of jobs carry location text too ambiguous to classify, limiting any location-based filter/map feature today.

Top countries

Rank	Country	Jobs	%
1	United States	79,427	51.0%
2	Germany	12,167	7.8%
3	United Kingdom	10,305	6.6%
4	India	3,471	2.2%
5	Canada	3,436	2.2%
6	France	2,218	1.4%
7	China	1,435	0.9%
8	Australia	1,225	0.8%

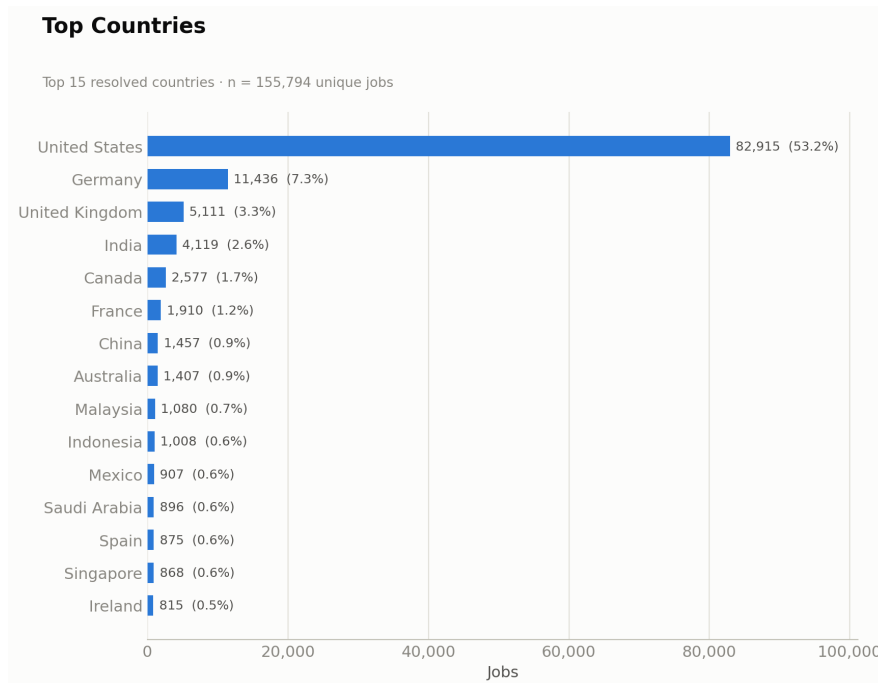


Figure 6. Top 15 resolved countries by job count.

Europe is largely a Germany story: 12,167 German jobs are 56% of the European total (two German public-agency sources). India, at 2.2% (3,471 jobs), is already the 4th-largest country despite no dedicated source (Section 10).

7. Job Freshness

Age is measured against the collection date (2026-07-03). A slim majority of the dataset was current at collection, but the stale fraction is already substantial and will only grow without a refresh mechanism.

Age at collection	Jobs	Cumulative 'within'	Cumulative %
Within 24 hours	24,052	24,052	15.4%
Within 3 days	+6,582	30,634	19.7%
Within 7 days	+11,884	42,518	27.3%
Within 14 days	+24,610	67,128	43.1%
Within 30 days	+16,949	84,077	54.0%
Older than 30 days	71,702	—	46.0%
Unknown / unparseable	15	—	0.0%

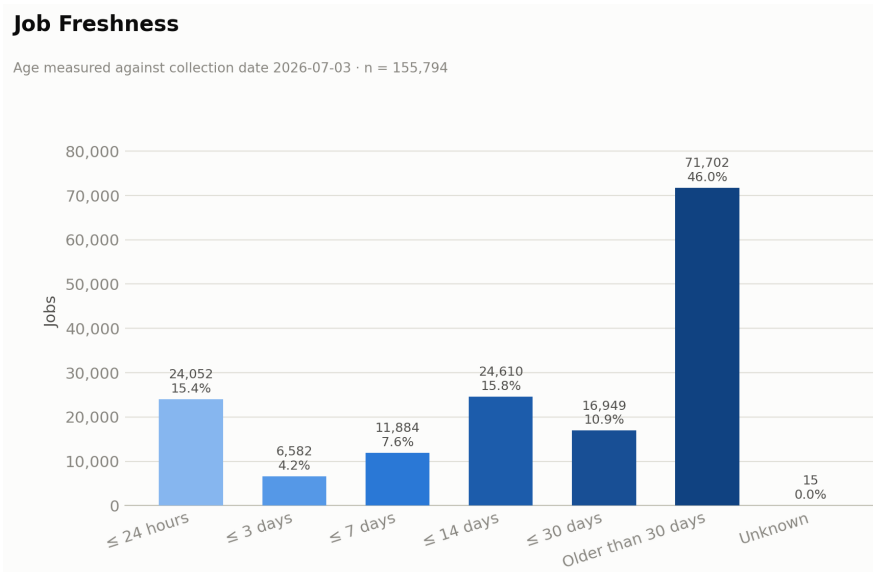


Figure 7. Job freshness distribution (buckets from freshest to stalest).

Total current jobs (≤ 30 days): 84,077 — 54.0% of the dataset. Total stale jobs (> 30 days, including unknown): 71,717 — 46.0%.

Deep-stale tail

Age	Jobs	%
Older than 90 days	45,075	28.9%
Older than 180 days	29,640	19.0%
Older than 365 days	21,768	14.0%

Oldest posting: 2009-12-05; median: 2026-06-09. The stale tail concentrates in large ATS boards that return every open requisition, including long-standing evergreen roles. With no refresh/expiry logic, the current/stale ratio degrades daily with no offsetting mechanism.

1. Current Active Jobs

Bucket	Jobs	% of dataset
Within 24 hours	24,052	15.4%
Within 3 days (cumulative)	30,634	19.7%
Within 7 days (cumulative)	42,518	27.3%
Within 14 days (cumulative)	67,128	43.1%
Within 30 days (cumulative)	84,077	54.0%
Active jobs (≤ 30 days)	84,077	54.0%
Potentially stale jobs (> 30 days, incl. unknown)	71,717	46.0%

Just over half the platform (54.0%) reads as active to a user; the remaining 46.0% (71,717 jobs) is old enough to question. Risk concentrates deeper in the tail: 28.9% of the whole dataset is >90 days old and 14.0% is >1 year old — the most urgent revalidation candidates.

2. Provider Freshness Ranking

Every provider, ranked by freshness rate (jobs ≤30 days ÷ provider total). The four lowest-volume/metered sources (SerpApi, OpenWeb Ninja, CareerJet, TheirStack) route through third-party domains and are shown combined, as elsewhere in this report.

Provider	Total Jobs	Jobs ≤30d	Freshness %	Notes
Bundesagentur	7,800	7,800	100.0%	Public agency feed; newest slice pulled each run

Provider	Total Jobs	Jobs ≤30d	Freshness %	Notes
Himalayas	2,964	2,964	100.0%	Newest-first pull of a 90,000+ job archive
ArbeItnow	930	930	100.0%	Small board fetched to full exhaustion every run
RemoteOK	100	100	100.0%	Fixed ~100-job newest-only snapshot; no pagination
Jobicy	100	100	100.0%	Fixed ~100-job snapshot; no pagination
Reed	9,000	8,658	96.2%	UK board with genuinely high listing turnover
Jooble	3,127	2,973	95.1%	Aggregator; surfaced listings turn over quickly
Adzuna	2,001	1,467	73.3%	Single-country (US) browse; moderate turnover
USAJOBS	10,000	6,138	61.4%	Federal postings often have long application windows
SmartRecruiters	65,193	36,479	56.0%	Mix of very-high-turnover and slower corporate boards
Other (SerpApi, OpenWeb Ninja, CareerJet, TheirStack, misc.)	10,108	5,091	50.4%	Mixed metered/aggregator sources
Teamtaylor	456	176	38.6%	Small company career-site RSS feeds
Greenhouse	14,930	5,013	33.6%	Full-board ATS fetch; many long-open requisitions
Ashby	5,171	1,688	32.6%	Full-board ATS fetch; slower-cycling roles
The Muse	2,000	615	30.8%	Curated aggregator; unauth browse skews older
Workable	4,159	1,232	29.6%	Full-board ATS fetch; longer time-to-fill boards
Lever	17,755	2,653	14.9%	Full-board ATS fetch; largest evergreen-posting share
TheirStack	0	0	n/a	Plan-gated (HTTP 402); contributes no jobs currently

The five 100%-fresh providers are either a bounded newest-first pull (Himalayas, Bundesagentur), a fixed ~100-newest snapshot (RemoteOK, Jobicy), or a small board fetched to exhaustion (ArbeItnow). Lever, Workable, The Muse, Ashby, and Greenhouse are full-board ATS fetches with no date filter — their freshness depends entirely on how long employers leave roles open, not on anything this pipeline controls.

3. Stale Job Analysis

Rank	Provider	Stale Jobs	Likely cause
1	SmartRecruiters	28,714	Evergreen ATS jobs — Domino's contributes tens of thousands of standing, store-level requisitions
2	Lever	15,102	Evergreen ATS jobs — full-board fetch returns every open corporate requisition regardless of age
3	Greenhouse	9,917	Evergreen ATS jobs — same full-board dynamic; many roles stay open for months
4	Other (SerpApi, OpenWeb Ninja, CareerJet, TheirStack, misc.)	5,017	Aggregated listings — re-surface third-party postings whose true origin date predates capture
5	USAJOBS	3,862	Government jobs — federal postings carry long, policy-mandated application windows
6	Ashby	3,483	Evergreen ATS jobs — growth-stage boards, slower-cycling roles
7	Workable	2,927	Evergreen ATS jobs — SMB/staffing boards with longer time-to-fill
8	The Muse	1,385	Slow provider updates — curated aggregator skewing toward already-listed jobs
9	Adzuna	534	Moderate — single-country browse, some longer-lived listings
10	Reed	342	Minimal — high-turnover UK board
11	Teamtaylor	280	Minimal — small boards, moderate turnover
12	Jooble	154	Minimal — aggregator with fast-turning listings
—	Bundesagentur, Himalayas, ArbeItnow, RemoteOK, Jobicy	0 each	100% fresh

SmartRecruiters, Lever, and Greenhouse together contribute 53,733 stale jobs (74.9% of all stale jobs) — a full-board-fetch artifact, not a defect. USAJOBS is stale for a legitimate reason (long application windows); aggregated listings are stale because they re-surface postings with an origin date this pipeline doesn't control.

4. Expired Job Detection

For every provider: does the API expose an expiry/closing date, a genuine last-updated timestamp, and can inactive/deleted postings be detected today?

Provider	Expiry date?	Closing date?	Last updated?	Inactive detectable?	Deleted identifiable?
Arbeitsw	No	No	No (created_at only)	No	No — needs run-to-run diffing
Himalayas	No	No	No (pubDate only)	No	No — needs run-to-run diffing
RemoteOK	No	No	No (epoch only)	No	No — snapshot-only API
Jobicy	No	No	No (pubDate only)	No	No — needs run-to-run diffing
The Muse	No	No	No (publication_date only)	No	No — needs run-to-run diffing
Bundesagentur	No	No	Partial (updates on republish)	Partial (age-filter proxy)	No — needs run-to-run diffing
Jooble	No	No	Yes (updated field, genuine)	Partial (date filter proxy)	No — needs run-to-run diffing
USAJOBS	Yes (in API, not yet extracted)	Yes (same field)	No (posting date only)	Yes, if extracted	Partial — drops from fresh search after close
Adzuna	No	No	No (created only)	Partial (max_days_old proxy)	No — needs run-to-run diffing
Reed	No	No	No (date only)	No	No — also hits an error boundary near pagination tail
SerpApi	No	No	No (relative text only)	No	No — Google Jobs exposes no closure signal
OpenWeb Ninja	No	No	No (posted timestamp only)	Partial (date_posted proxy)	No — needs run-to-run diffing
CareerJet	No	No	No (date only)	No	No — needs run-to-run diffing
TheirStack	No	No	No (date_posted only)	Partial (age-filter proxy)	No — also currently plan-gated (0 jobs)
Greenhouse	No	No	Yes (updated_at, already used as fallback)	No native field	Yes — full board each run, absence-based diffing reliable
Lever	No	No	No (createdAt only)	No native field	Yes — full board each run, absence-based diffing reliable
Ashby	No	No	No (publishedAt only)	No native field	Yes — full board each run, absence-based diffing reliable
SmartRecruiters	No	No	No (releasedDate only)	No native field	Yes — full board each run (care needed at Domino's scale)
Workable	No	No	No (published_on/created_at only)	No native field	Yes — full board each run, absence-based diffing reliable
Teamtailor	No	No	No (pubDate only)	No native field	Yes — RSS returns full current list each run

Almost no provider exposes a native expiry/closing date — only USAJOBS has one (ApplicationCloseDate), and it isn't yet extracted. Only Jooble (updated) and Greenhouse (updated_at) offer a genuine last-updated signal. The six full-board ATS providers can't detect inactivity, but can detect deletions via run-to-run absence diffing — the missing piece is persistence, not API capability.

5. Freshness Improvement Strategy

- Incremental synchronization: persist a per-job first_seen/last_seen watermark (using createdAt/publishedAt/releasedDate/updated_at, already returned by every provider) so each run can tell new postings from ones already known.
- Provider-specific refresh schedules: match cadence to each provider's own turnover rate rather than one schedule for all 20.

- Missing-job detection: for the six full-board ATS providers, compare each run's job URLs/IDs against the previous run's; anything present last time but absent now has closed and should be flagged or removed.
- Automatic expiration: for the 16 providers with no closure signal at all, apply an age-based soft-expiry policy — flag or hide listings past a configurable threshold (60-90 days) as likely inactive even without a hard signal.
- Run-to-run comparison: the mechanism underlying both missing-job detection and automatic expiration — requires moving from the current single-file overwrite model to a persisted, keyed history of what was seen on each run.
- Automatic stale-job removal: once run-to-run comparison exists, jobs absent for N consecutive runs (full-board ATS) or past a soft-expiry threshold (no-signal providers) can be automatically retired rather than accumulating indefinitely.
- Daily vs. weekly synchronization: daily for SmartRecruiters, Reed, Bundesagentur, USAJOBS, and Jooble (high-freshness-rate, high-volume providers already turning over quickly); weekly or less for Lever, Workable, The Muse, Ashby, Greenhouse, and Teamtailor, where daily re-fetching mostly re-downloads the same long-open postings — effort is better spent on incremental diffing for these than on fetch frequency alone.

6. Management Summary

- What percentage of jobs are likely active today? 54.0% (84,077 of 155,794) were posted within the last 30 days and would read as current to a typical user.
- What percentage should be revalidated? 46.0% (71,717 jobs) are over 30 days old; the highest-priority subset is the 28.9% of the entire dataset older than 90 days, and especially the 14.0% older than a full year, where the risk of showing an already-closed position is greatest.
- Which providers need the most improvement? Lever (14.9% fresh), Workable (29.6%), The Muse (30.8%), Ashby (32.6%), and Greenhouse (33.6%) have the lowest freshness rates; SmartRecruiters, despite a middling 56.0% rate, contributes the single largest absolute number of stale jobs (28,714) and should be prioritized for that reason.
- What engineering work will improve freshness the most? Incremental synchronization with run-to-run comparison, applied first to the three highest-absolute-stale-volume providers (SmartRecruiters, Lever, Greenhouse — together 74.9% of all stale jobs), combined with turning on the date filters already available but unused on Adzuna, USAJOBS, Bundesagentur, Jooble, and TheirStack — the same conclusion reached independently in research/api_analysis.md (Sections 17 and 20), requiring no new provider integration.

8. Duplicate Analysis

The pipeline deduplicates by exact URL — zero duplicate and zero empty URLs remain. Under looser candidate keys, many same-title/company rows appear to “collide”; this section quantifies that and shows it is not a duplication problem.

Check	Result
Exact-URL duplicates remaining	0
Jobs with empty URL	0
Title + company collisions (extra rows)	35,300 across 11,340 groups
Title + company + location collisions (extra rows)	8,290 across 4,963 groups
Title+company groups spanning more than one distinct URL	11,340 (100% of those groups)

Duplicate Analysis

*Not true duplicates — same-title/company rows at different URLs, overwhelmingly legitimate multi-location postings (see report).

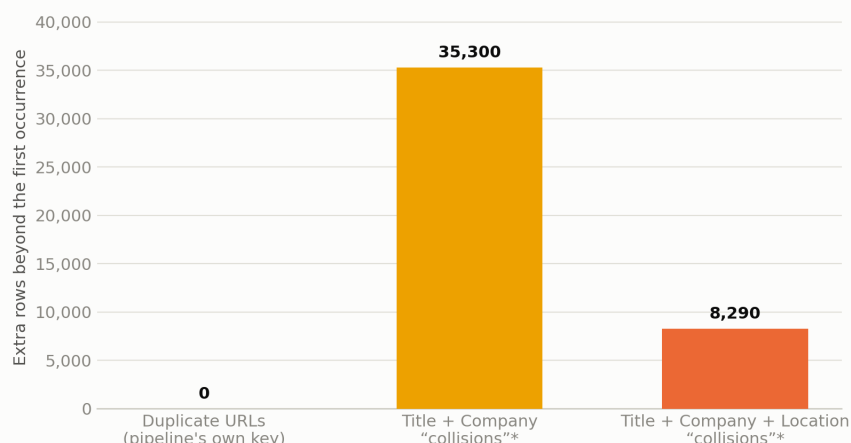


Figure 8. Duplicate counts under three candidate keys.

Every one of the 11,340 title+company groups spans multiple distinct URLs — not missed duplicates, but the same role posted at hundreds of separate locations (Domino's "Delivery Driver," BoxLunch "Sales Associate"). A naive title+company dedup rule would have erased tens of thousands of legitimate openings.

The real, smaller duplicate risk is cross-provider overlap (Jooble/Careerjet/Adzuna occasionally re-listing the same posting) — not measured by URL dedup today. Any future effort should target this conservatively, never by collapsing on title+company alone.

9. API Analysis

All 20 providers were audited for auth model, rate/result-window limits, pagination, and filters. Three already hit a hard result-window cap (USAJOBS/Bundesagentur 10,000; Reed ~9,900). Three more are deliberately throttled on metered/paid APIs (SerpApi, OpenWeb Ninja, TheirStack, ~50/run).

Priorit y	Providers	Rationale
Critical	SmartRecruiters, Adzuna	SmartRecruiters is already the volume backbone; Adzuna is the single highest-leverage untapped asset (one keyed integration can add several regions by iterating its country parameter).
High	Lever, Greenhouse, USAJOBS, Reed, Bundesagentur	Large, free/near-free backbones for major regions, though each is bounded by a curated company list, federal-only scope, or a hard result window.
Medium	Ashby, Workable, Himalayas, The Muse, Jooble, Careerjet	Solid contributors with real breadth but individually smaller or with untapped upside (e.g., Himalayas' richer search endpoint).
Low	Arbeitsnow, RemoteOK, Jobicy, Teamtailor, SerpApi, OpenWeb Ninja, TheirStack	Small absolute volume or deliberately throttled metered sources; TheirStack currently contributes zero jobs.

Full per-provider detail — authentication, pricing, rate limits, pagination, and every supported filter — is documented in research/api_analysis.md, Sections 4 and 9.

10. India Research

India is already the 4th-largest country in the dataset (3,471 jobs, 2.2%) with no dedicated source. A survey of the Indian recruitment ecosystem (government portals, private boards, ATS/HR platforms, staffing firms) found a consistent pattern: a large, active hiring market with almost no aggregatable public job APIs.

Key findings

- Government: the National Career Service (NCS) publishes vacancy data via data.gov.in as a periodic open dataset, not a live searchable API — authoritative but freshness-limited.

- Private boards: Naukri, Foundit, Apna, Shine, TimesJobs, Instahyre, Hirst, iimjobs, Internshala, Unstop, Wellfound, and LinkedIn/Indeed India all expose no official public read API.
- India-origin ATS/HRMS platforms (Zoho Recruit, Freshteam, Darwinbox, Keka, GreytHR) are uniformly per-tenant — useful only to that specific employer's own integration, not aggregatable across employers.
- Major Indian staffing firms (TeamLease, Qess, Randstad India, etc.) expose no public job feeds.

Recommended routes (ranked)

Rank	Route	Effort	ROI
1	ATS slug expansion for Indian-HQ companies on Greenhouse/Lever/Ashby/Workable/SmartRecruiters (already integrated)	Low	High
2	India-scoped query on existing metered sources (JSearch/OpenWeb Ninja, SerpApi)	Low	High
3	Paid aggregator (Fantastic Jobs / Active Jobs DB) covering Indian ATS and boards	Low	Medium (budget-gated)
4	National Career Service open dataset (data.gov.in)	Medium (ETL)	Medium
5	Enterprise data providers (Coresignal, JobsPikr)	Medium	Low (cost)
—	Scraping Naukri / Foundit / Apna / LinkedIn / Indeed	—	Not recommended (legal/maintenance risk)

Bottom line: legal, low-maintenance India coverage comes from extending infrastructure already built for this project (ATS slugs, existing metered/Google-Jobs sources) — not a new Indian ATS vendor or scraping.

India API Landscape & Coverage Strategy

This subsection expands the India research above into a management-focused view: why India has fewer public job APIs than the US or Europe, and what practical, low-risk options exist to grow Indian coverage using infrastructure this project already runs. All figures are drawn from research/api_analysis.md (Sections 11 and 16); nothing here is a new estimate.

1. Indian Job Platform Landscape

Major hiring platforms serving the Indian market, compared against the six ATS providers already integrated in this project, on the attributes that determine whether each is a realistic aggregation source.

Platform	Public API	Enterprise API	Govt	ATS	Auth Req.	Free	Paid	Region Covered	Typical Companies	Recommendation
Naukri	No	No (internal v2, IP-gated)	No	No	N/A (no public API)	No	No	India	Broad Indian employers (1M+ jobs)	Not recommended - no public API; internal endpoint is a ToS grey area
Foundit (Monster India)	No	No	No	No	N/A (no public API)	No	No	India	Broad Indian employers (800k+ jobs)	Not recommended - no public API, scraping only
Apna	No	No	No	No	N/A (no public API)	No	No	India	Blue-collar / SMB hiring	Not recommended - no public API
LinkedIn Jobs	No	Yes (partner OAuth)	No	No	Partner OAuth	No	Yes (partner only)	Global incl. India	All company sizes	Not recommended - no open jobs API for a project like this
Indeed India	No (publisher API deprecated)	Yes (partner/commercial)	No	No	Partner	No	Yes (partner only)	Global incl. India	All company sizes	Not recommended - consumer search API deprecated, partner-only
National Career Service (NCS)	No (open dataset, not a live API)	No	Yes	No	None (dataset); OGD key for some resources	Yes	No	India (national)	Government vacancies	Investigate - authoritative breadth, poor freshness (periodic dataset)

Platform	Public API	Enterprise API	Govt.	ATS	Auth Req.	Free	Paid	Region Covered	Typical Companies	Recommendation
Greenhouse	Yes (per-company board)	No	No	Yes (already integrated)	None	Yes	No	Global - incl. Indian unicorns/GCCs	Razorpay, Freshworks, and other India-HQ/India-office companies	Add India-HQ slugs - highest-ROI India route
Lever	Yes (per-company)	No	No	Yes (already integrated)	None	Yes	No	Global - incl. Indian companies	India-HQ and India-office tech companies	Add India-HQ slugs
SmartRecruiters	Yes (postings API)	No	No	Yes (already integrated)	None	Yes	No	Global - incl. Indian companies	Large enterprises with India operations	Add India-HQ slugs
Ashby	Yes (per-company)	No	No	Yes (already integrated)	None	Yes	No	Global - AI/SaaS-heavy, incl. Indian startups	India-HQ AI/SaaS startups	Add India-HQ slugs
Workable	Yes (widget API)	No	No	Yes (already integrated)	None	Yes	No	Global - incl. Indian SMB/staffing	India-HQ SMB/consulting companies	Add India-HQ slugs
Teamtaylor	Yes (RSS per company)	No	No	Yes (already integrated)	None	Yes	No	Global - smaller India presence	India-HQ companies using Teamtailor	Add India-HQ slugs (smallest expected yield)

2. Why India Has Fewer Public APIs

- Most large Indian job boards (Naukri, Foundit, Apna) are commercial platforms that monetize through employer subscriptions, not open data-sharing - there is no business incentive to expose a free public read API.
- Public job-search APIs are generally not provided: the closest thing (Naukri's internal v2 endpoint) is undocumented, India-IP-gated, and a ToS grey area rather than a real integration path.
- Enterprise partnerships are the norm for bulk data access instead - LinkedIn and Indeed both offer only partner/OAuth-gated commercial access, not a public developer API.
- ATS platforms expose public endpoints because companies host their own job boards on that infrastructure: Greenhouse, Lever, Ashby, Workable, SmartRecruiters, and Teamtailor all provide a no-auth, per-company feed precisely because the employer - not the platform - owns and publishes that board; India-origin ATS/HRMS platforms (Zoho Recruit, Freshteam, Darwinbox, Keka, GreytHR) do not follow this pattern and expose only per-tenant APIs gated behind each employer's own credentials.

3. Best Strategy for Increasing Indian Coverage

The nine candidate approaches, ranked by expected impact. Additional-jobs figures are planning estimates consistent with this report's existing per-provider optimization ranges (Section 17), not new figures computed from jobs.json - India-HQ slugs do not yet exist in the curated company lists.

Strategy	Engineering Effort	Expected Additional Jobs	Maintenance	Priority
Expand Greenhouse companies (India-HQ slugs)	Low	Hundreds to low thousands	Low (slug curation)	High
Expand SmartRecruiters companies (India-HQ slugs)	Low	Hundreds to low thousands	Low (slug curation)	High
Expand Lever companies (India-HQ slugs)	Low	Hundreds to low thousands	Low (slug curation)	High
Expand Ashby companies (India-HQ slugs)	Low	Hundreds to low thousands	Low (slug curation)	High
Expand Workable companies (India-HQ slugs)	Low	Low hundreds	Low (slug curation)	Medium-High
Expand Teamtailor companies (India-HQ slugs)	Low	Low hundreds	Low (slug curation)	Medium

Strategy	Engineering Effort	Expected Additional Jobs	Maintenance	Priority
India-specific queries on existing APIs (JSearch/OpenWeb Ninja, SerpApi)	Low (reuses existing code)	Modest, budget-bound (metered quotas)	Low	High
National Career Service open dataset (data.gov.in)	Medium (bulk ETL)	Large in aggregate	Medium-High (periodic re-ETL)	Medium
Enterprise partnerships with Indian job boards (Naukri, Foundit, etc.)	High (commercial negotiation)	Very large if secured	Medium-High (contract-dependent)	Low-Medium (investigate later)

The six ATS-expansion rows together are estimated at 5,000-20,000 additional India jobs (research/api_analysis.md Sections 10.8 and 11.3) - by far the best free, legal, low-maintenance route, and the same conclusion this report's India Research section above already reaches independently.

4. Indian ATS Opportunity

Many Indian startups, unicorns, Global Capability Centers (GCCs), and multinational companies already run their careers pages on the same six ATS platforms this project integrates - Razorpay, PhonePe, CRED, Groww, Meesho, BrowserStack, Freshworks, and Postman are documented examples (research/api_analysis.md Section 11.3). None of these require a new provider integration: each is reachable the same way any other Greenhouse, Lever, Ashby, Workable, SmartRecruiters, or Teamtailor company already is - by adding a verified company slug to the existing curated list.

Expanding the monitored ATS company list toward these India-HQ companies is therefore likely to provide significantly more Indian jobs without integrating a single new API - the same pattern already used to grow every other region in this project.

5. Management Summary

- India currently has fewer publicly available job APIs than the US or Europe - nearly every major Indian consumer job board exposes no official public API.
- The biggest opportunity is expanding ATS company coverage (Greenhouse, Lever, Ashby, Workable, SmartRecruiters, Teamtailor) toward India-HQ companies, rather than searching for additional public APIs that largely do not exist for this market.
- Enterprise APIs and paid aggregators (LinkedIn, Indeed, Naukri partnerships, or commercial data providers) can be investigated later if broader commercial coverage becomes necessary beyond what ATS expansion provides.

11. Additional Regional APIs

A verified market survey covered Canada, the UK, Europe, Singapore, Southeast Asia, and LATAM for additional free/public sources beyond the 20 already integrated.

Region	Best candidate	Effort	Priority
Canada	Adzuna 'ca' (existing key, config-level change)	Trivial	Critical
United Kingdom	Adzuna 'gb' (de-risks single-source dependency on Reed)	Trivial	High
Europe	Adzuna multi-country (fr, de, es, nl, it, at, pl)	Trivial	Critical
Europe	France Travail (official, free, OAuth2 registration)	Medium	High
Singapore	MyCareersFuture — official, no-auth public API (~80k jobs)	Low	Critical
Southeast Asia	ATS slug expansion for SEA-headquartered companies	Low	Medium
LATAM	Adzuna 'br' + 'mx' (existing key)	Trivial	Critical
LATAM	Get on Board — the one LATAM board with a real public API	Low-Medium	Investigate

20 well-known regional boards were explicitly ruled out — Eluta/Workopolis/Jobillico/Talent.com/WowJobs (Canada); TotalJobs/CV-Library/Jobsite/CWJobs (UK); StepStone/Xing Jobs/Jobrapido/Welcome to the Jungle (Europe); Computrabajo/Bumeran/OCC Mundial (LATAM); JobStreet/Jobsonline/Kalibrr/Glints (SEA) — each closed or paid-partner-only. Get on Board (LATAM) was the sole exception with a genuine public API.

Headline finding: the Adzuna key already held queries only the US today. Extending it to more countries is by far the single highest-leverage change available — it touches five of six target regions with no new integration work.

12. API Filter Optimization

An audit of all 20 integrations found most run deliberately unfiltered “browse everything” queries — right for a first pass, but leaving three significant levers unused.

- Date filters (e.g., Adzuna max_days_old, USAJOBS DatePosted, Jooble datecreatedfrom) are supported by at least six integrated providers but used by none — turning them on would cut the stale tail at ingest rather than after the fact.
- Country/locale filters (Adzuna's country path, Careerjet's locale_code, Himalayas' country parameter) are supported but mostly unused — turning them on directly targets the weak regions identified in Section 6.
- Query partitioning — splitting one broad query into several filtered ones by category, location, or date range — is the only way to exceed the hard result-window caps already reached by USAJOBS (10,000), Bundesagentur (10,000), Reed (~9,900), The Muse (2,000), Jooble (~1,100), and Himalayas (3,000 of a 90,000-job archive). This is the single largest coverage lever available without adding any new provider.

Planning estimate: turning on these filters and partitioning the capped sources could add ~40,000–90,000 useful, fresher, less-duplicated jobs with zero new providers (full per-provider detail: [research/api_analysis.md §17](#)).

13. Maximum Coverage from Existing Providers

Every provider was audited individually on a narrower question: is it already returning everything realistically obtainable from its own API's rules — and if not, how much more is available using only capabilities that API already exposes?

Are we already extracting the maximum realistic number of jobs from each provider?

For 17 of 20 providers, no — meaningful additional volume is available from capabilities the API already exposes, no new auth/scraping/contracts required. Only Arbeitnow, RemoteOK, and TheirStack are already at their ceiling (TheirStack's is a billing restriction, not an API limit).

Which providers still have untapped potential?

- Adzuna — by far the largest opportunity in the entire provider set. The current 2,000-job figure reflects querying a single country (the US) out of 10+ countries the existing API key already has access to.
- Bundesagentur, USAJOBS, and Reed — each already sits at or near a hard per-query result-window ceiling (10,000, 10,000, and ~9,900 respectively), but each also supports a location or keyword filter that can open an independent window of similar size when the query is partitioned instead of run once, broadly.
- Himalayas, The Muse, Jooble, and Careerjet — each is currently run as a single broad or wildcard query well below its real ceiling; switching to filtered, partitioned queries (by country, category, or real keywords instead of a wildcard) would meaningfully increase volume.

Which providers have already reached their practical limits?

- Arbeitnow — the entire current board is already fetched to exhaustion via full pagination, and the API offers no keyword, location, or category parameter to partition further.
- RemoteOK — the documented public endpoint is a fixed ~100-job snapshot with no confirmed filtering or pagination mechanism.
- TheirStack — effectively at zero due to an account plan restriction (HTTP 402), not because its API has been exhausted; this should be reassessed the moment Jobs API access is restored on the account.

Which providers can be improved using existing API filters?

Eight providers (Adzuna, Bundesagentur, USAJOBS, Reed, Himalayas, Jooble, Careerjet, The Muse) already support unused date/country/category/location filters. Turning these on is a pure configuration change — no new credentials required.

Which providers require adding additional companies rather than changing code?

All six ATS providers expose no request filter at all — each is a full-board, per-company fetch already fully exploited for its configured companies. The entire opportunity is expanding the curated company-slug list, not the query.

Which providers cannot realistically return more jobs due to API limitations?

Arbeitnow and RemoteOK — both expose a complete, unfilterable snapshot with no query dimension to expand. Further volume depends on the boards themselves growing, not on anything this project controls.

Provider-by-provider comparison

Provider	Current Jobs Collected	Estimated Practical Maximum	Additional Jobs Possible	Recommended Optimization Strategy
Arbeitnow	930	~950–1,000	~0–100	None — full board already exhausted
Himalayas	3,000	13,000–23,000	+10,000–20,000	Switch to richer search endpoint; partition by country/category
RemoteOK	100	~100	~0	None confirmed; verify tag-scoped endpoints before assuming more
Jobicy	100	300–500	+200–400	Partition by geo/industry/tag across multiple calls
The Muse	2,000	5,000–8,000	+3,000–6,000	Partition by category/location; consider an API key
Bundesagentur	10,000	25,000–40,000	+15,000–30,000	Partition by region or keyword
Jooble	1,129	4,000–9,000	+3,000–8,000	Replace wildcard with targeted keywords/locations
USAJOBS	10,000	20,000–30,000	+10,000–20,000	Partition by job category or location
Adzuna	2,000	22,000–42,000	+20,000–40,000	Query additional countries already covered by the existing key
Reed	9,000	17,000–24,000	+8,000–15,000	Partition by UK region/keyword
SerpApi	50	~2,500/mo (budget-bound)	Budget-dependent	Reallocate quota to targeted location/date queries
OpenWeb Ninja	~47	~200/mo (budget-bound)	Budget-dependent	Reallocate quota toward country-scoped (e.g. India) queries
Careerjet	2,000	7,000–12,000	+5,000–10,000	Add locale-scoped passes + real keywords
TheirStack	0	0 (plan-gated)	0 until plan restored	Resolve the account/billing restriction first
Greenhouse	22,728	27,728–37,728	+5,000–15,000	Expand curated company-slug list
Lever	17,755	22,755–27,755	+5,000–10,000	Expand curated company-slug list
Ashby	5,171	7,171–10,171	+2,000–5,000	Source additional slugs outside AI/SaaS/FinTech
SmartRecruiters	65,193	70,193–80,193	+5,000–15,000	Expand slug list, but weigh against concentration risk
Workable	7,644	10,644–14,644	+3,000–7,000	Source slugs in staffing/consulting/fintech/healthcare
Teamtailor	474	974–1,974	+500–1,500	Source additional career sites (highest candidate hit rate)

Summary

- Highest ROI first: Adzuna (+20,000–40,000, additional countries on the existing key), then Bundesagentur, USAJOBS, Reed, Himalayas, Jooble, Careerjet — together ~56,000–99,000 more via filters/partitioning already supported.
- Already near maximum: Arbeitnow and RemoteOK (no query-based lever left); TheirStack is at zero due to billing, not an API ceiling.
- Expected increase before any new API: ~75,000–140,000 additional jobs from the eight already-credentialed providers' untapped filter capacity alone.

14. Recent Job Market Analysis (≤30 Days)

Business question: what would a user see on the platform today? Figures below cover the 84,077 jobs posted within 30 days of collection (54.0% of the dataset); older jobs are excluded except where noted.

1. Current Job Categories

Technology expands into 11 sub-categories, Design into 3, Business into 15. This dataset skews heavily retail/hospitality/food-service/logistics, so a majority of recent jobs fall outside this taxonomy and are honestly reported as Other rather than forced into a poor fit.

Category	Recent Jobs	% of recent	Group
Other	44,842	53.3%	Unclassified
Supply Chain	7,503	8.9%	Business
Customer Support	5,251	6.2%	Business
Software Engineering	5,225	6.2%	Technology
Healthcare	3,482	4.1%	Other named
Sales	2,770	3.3%	Business
Executive	1,568	1.9%	Other named
Education	1,465	1.7%	Other named
Accounting	1,221	1.5%	Business
Project Management	986	1.2%	Business
Internship / Graduate	911	1.1%	Other named
Human Resources	773	0.9%	Business
Marketing	736	0.9%	Business
Manufacturing	639	0.8%	Other named
Product Management	565	0.7%	Business
Finance	522	0.6%	Business
Data Science	511	0.6%	Technology
Legal	399	0.5%	Other named
Business Development	392	0.5%	Business
Cybersecurity	374	0.4%	Technology
Administrative	334	0.4%	Other named
AI / Machine Learning	299	0.4%	Technology
QA / Testing	292	0.3%	Technology
Digital Marketing	291	0.3%	Business
Customer Success	289	0.3%	Business
DevOps	276	0.3%	Technology
Operations	273	0.3%	Business
Cloud	268	0.3%	Technology
Procurement	257	0.3%	Business
Recruiting	247	0.3%	Business
Full Stack	203	0.2%	Technology
Backend	197	0.2%	Technology
IT Support	169	0.2%	Other named
UI / UX	100	0.1%	Design
Product Design	99	0.1%	Design
Government	92	0.1%	Other named
Pharmaceutical	77	0.1%	Other named
Graphic Design	71	0.1%	Design
Frontend	61	0.1%	Technology
Mobile	47	0.1%	Technology

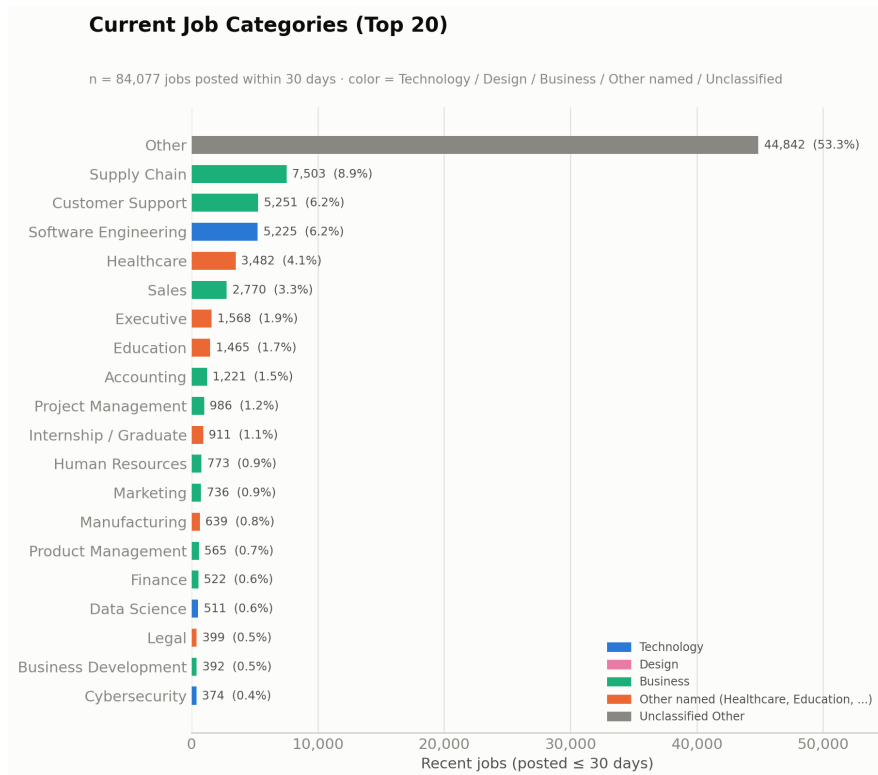


Figure 13. Current job categories (top 20), color-coded by group.

Macro mix: Technology = 7,753 jobs (9.2%), Design = 270 jobs (0.3%), Business = 22,076 jobs (26.3%), other named categories (Healthcare, Education, Manufacturing, etc.) = 9,136 jobs (10.9%), Unclassified Other = 44,842 jobs (53.3%).

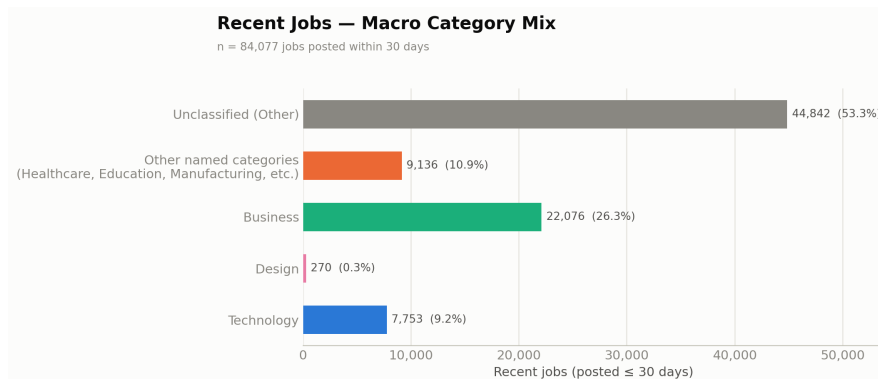


Figure 14. Recent jobs, macro category mix.

Excluding Other: Supply Chain (8.9%), Customer Support (6.2%), and Software Engineering (6.2%) lead, followed by Healthcare (4.1%) and Sales (3.3%). Within Technology, Software Engineering dominates; Mobile and Frontend (0.1% each) are smallest.

2. Current Jobs by Region

Region	Recent Jobs	% of recent
United States	39,124	46.5%
Europe	15,721	18.7%
Other	14,372	17.1%
United Kingdom	6,925	8.2%
LATAM	2,886	3.4%
Singapore / Southeast Asia	1,744	2.1%

Region	Recent Jobs	% of recent
India	1,656	2.0%
Canada	1,649	2.0%

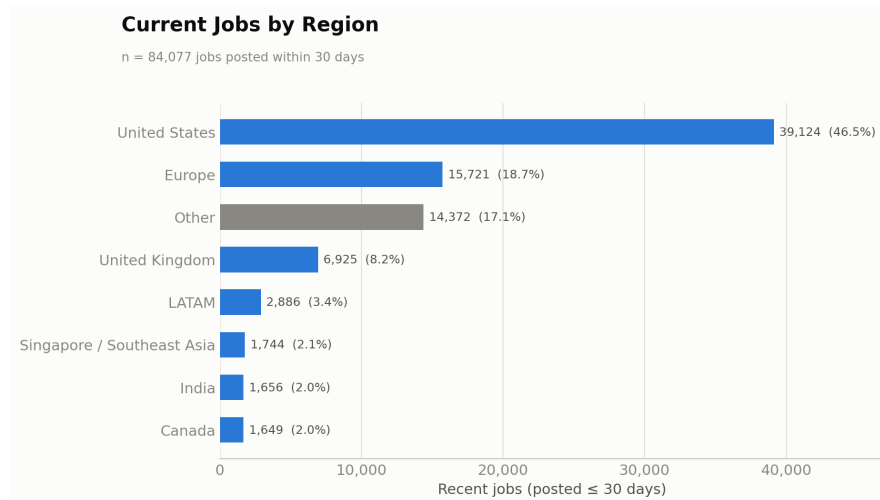


Figure 15. Current jobs by region.

The US (46.5%) and Europe (18.7%) account for nearly two-thirds of recent postings; the UK (8.2%) is third. Canada, India, and Singapore/SEA are each near 2% — the weakest activity, consistent with the coverage gaps in research/api_analysis.md.

3. Current Jobs by Provider

Provider	Recent Jobs	% of provider's own jobs that are recent
SmartRecruiters	36,479	56.0%
Reed	8,658	96.2%
Bundesagentur	7,800	100.0%
USAJOBS	6,138	61.4%
Other (SerpApi, OpenWeb Ninja, CareerJet, TheirStack, misc.)	5,091	50.4%
Greenhouse	5,013	33.6%
Jooble	2,973	95.1%
Himalayas	2,964	100.0%
Lever	2,653	14.9%
Ashby	1,688	32.6%
Adzuna	1,467	73.3%
Workable	1,232	29.6%
Arbeitsnow	930	100.0%
The Muse	615	30.8%
Teamtailor	176	38.6%
Jobicy	100	100.0%
RemoteOK	100	100.0%

Bundesagentur, Himalayas, Arbeitsnow, RemoteOK, and Jobicy are 100% fresh; Reed (96.2%) and Jooble (95.1%) follow. Lever (14.9%), Workable (29.6%), The Muse (30.8%), Ashby (32.6%), and Greenhouse (33.6%) are lowest — full-board ATS fetches with no date filter (see Section 7).

4. Current Hiring Companies

Rank	Company	Recent Jobs	% of recent
1	Domino's	20,620	24.5%

Rank	Company	Recent Jobs	% of recent
2	AccorHotel	3,397	4.0%
3	AECOM	2,297	2.7%
4	Bosch Group	1,900	2.3%
5	Veterans Health Administration	1,675	2.0%
6	Anduril Industries	817	1.0%
7	Northwestern Memorial Healthcare	751	0.9%
8	Reed	689	0.8%
9	SpaceX	681	0.8%
10	Delivery Hero	599	0.7%
11	eFinancialCareers	514	0.6%
12	Primark	454	0.5%
13	ASSYSTEM	417	0.5%
14	Tesco	396	0.5%
15	AUREA GmbH	341	0.4%
16	Boyd Gaming	335	0.4%
17	Munson Healthcare	328	0.4%
18	NielsenIQ	324	0.4%
19	Hays Specialist Recruitment Limited	320	0.4%
20	Continental	309	0.4%

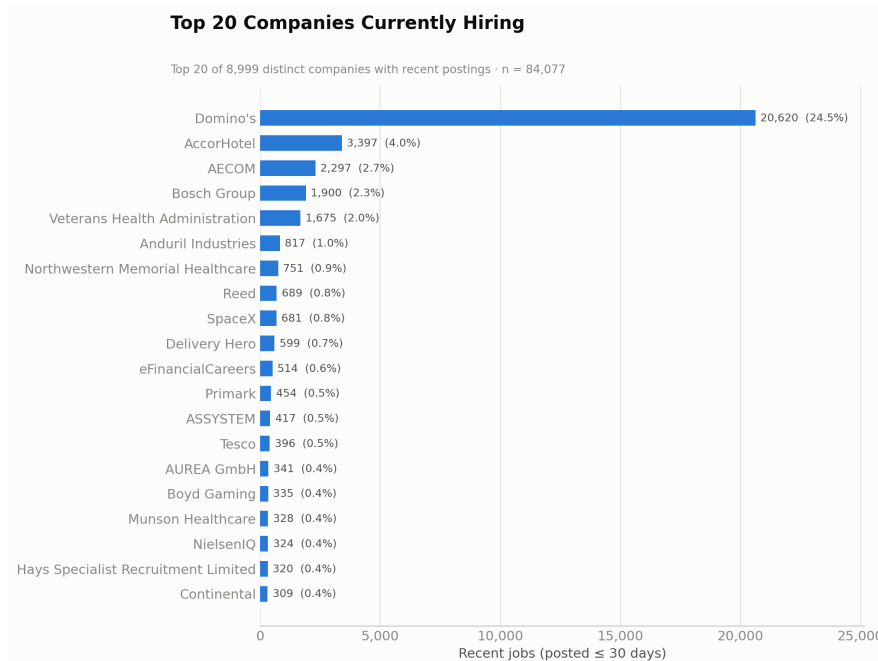


Figure 16. Top 20 companies currently hiring.

The top 10 companies are 39.8% of recent postings; Domino's alone is 24.5% (up from 15.7% dataset-wide) because its board refreshes unusually often. 8,999 distinct companies post recently, most with just one opening.

5. Current Hiring Trends

Methodology note: jobs.json is a single snapshot, so no true multi-run trend exists. "Growth" below proxies momentum by comparing each category's/region's last-7-day share vs. its 8–30-day share — a within-snapshot signal, not month-over-month.

Rank	Fastest-growing technical category	Jobs (<30d)	Share posted in last 7 days
1	Full Stack	203	55.7%

Rank	Fastest-growing technical category	Jobs (<30d)	Share posted in last 7 days
2	DevOps	276	53.3%
3	Cloud	268	47.0%
4	Mobile	47	46.8%
5	Backend	197	46.2%

Rank	Fastest-growing non-technical category	Jobs (<30d)	Share posted in last 7 days
1	Digital Marketing	291	64.9%
2	Accounting	1,221	55.7%
3	Human Resources	773	55.6%
4	Business Development	392	54.6%
5	Finance	522	52.7%

Remote share is lower among recent jobs (9.7%) than older ones (12.3%) — onsite/hybrid postings refresh faster here, not a sign remote hiring itself is shrinking.

Regional trend (recent share vs. dataset share): Europe +4.6pts (18.7% vs 14.1%) and UK +1.6pts (8.2% vs 6.6%) over-index recently; the US is -4.0pts (46.5% vs 50.5%) — still dominant but under-contributing. Canada/India/LATAM/SEA are each slightly below their overall share.

Top-20-employer industries driving current hiring: food service/QSR (Domino's), hospitality (AccorHotel, Boyd Gaming), engineering/construction (AECOM, ASSYSTEM), manufacturing (Bosch, Continental), healthcare (VHA, Northwestern Memorial, Munson), aerospace/defense (Anduril, SpaceX), and staffing (Reed, Hays).

6. Business Insights

A visitor today would mostly see retail, food-service, and logistics/delivery roles (Other + Supply Chain = 62.2% of recent postings), driven overwhelmingly by Domino's (24.5% of all recent jobs).

Dominant industries: food service/QSR, hospitality, healthcare, and engineering/industrial consulting.

Dominant technical skill: Software Engineering (6.2% of recent jobs, 67% of recent Tech). Data Science, Cybersecurity, and AI/ML are each under 0.7%; Technology overall is 9.2% of current listings.

Dominant non-technical roles: Supply Chain (8.9%) and Customer Support (6.2%), followed by Healthcare (4.1%) and Sales (3.3%).

Most opportunities by region: US (46.5%) and Europe (18.7%), UK a distant third (8.2%); Canada, India, and Singapore/SEA each offer comparatively few (~2%).

7. Engineering Recommendations

- Categories needing coverage: Technology (9.2%) and Design (0.3%) are thin, especially Mobile/Frontend/Cloud — growing ATS slug lists toward mobile/cloud/design employers would help.
- Regions needing providers: Canada, India, and Singapore/SEA (~2% each) — multi-country Adzuna, MyCareersFuture, and ATS slug expansion are the identified low-effort fixes.
- Refresh more frequently: SmartRecruiters, Reed, Bundesagentur, USAJOBS, Jooble — high-freshness, high-volume, already turning over fast enough to justify daily sync.
- Already sufficient: Bundesagentur, Himalayas, Arbeitnow, RemoteOK, Jobicy (100% fresh), plus Reed (96.2%) and Jooble (95.1%) — no freshness work needed.
- Optimize first: Lever (14.9%), Workable (29.6%), The Muse (30.8%), Ashby (32.6%), Greenhouse (33.6%) — large contributors disproportionately stale; since none expose a date filter, incremental sync is the fix.

15. Engineering Recommendations

- Broaden regional balance — particularly Canada, UK resilience, and Singapore/Southeast Asia — by extending already-integrated providers rather than adding new integrations.
- Introduce a scheduled refresh and expiry mechanism so the freshness ratio stops degrading between runs and closed postings are retired automatically.

- Grow ATS company-slug lists for under-represented regions (Canada, SEA, LATAM, India) to rebalance provider and regional concentration.
- Do not deduplicate on title+company alone — doing so would remove tens of thousands of legitimate multi-location jobs. Any future duplicate-reduction work should combine URL normalization with conservative, location-aware matching.
- Improve location-text normalization to shrink the 18.2% unclassifiable bucket and make regional filtering and reporting more reliable.
- Revisit the TheirStack account/plan to restore its contribution, or formally retire it from the active provider list if the restriction is permanent.
- Monitor concentration risk — with 41.8% of jobs from one provider and 15.7% from one company, any change to SmartRecruiters' Domino's board would materially shift dataset-wide statistics.
- Turn on already-supported date and country/locale filters (Section 12) before considering any new integration — this is free, low-risk, and high-impact.

16. Regional API Expansion Strategy

Table-format note: Type/Pricing consolidate the requested Public/Government/Commercial/ATS and Free/Trial/Paid/Enterprise flags into two categorical columns (every attribute still answered). Figures are drawn from research/api_analysis.md §§4,5,9,10,11,17,18,20 — nothing here is a new estimate.

United States — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
Adzuna (US)	Public/Commercial aggregator	Free tier (keyed)	Yes	Yes
USAJOBS	Government (US federal)	Free	Yes (3 headers)	Yes
Greenhouse/Lever/Ashby/SmartRecruiters/Workable/Teamtailor	ATS	Free (no auth)	No	Yes
Jooble/Careerjet/SerpApi/OpenWeb Ninja	Commercial aggregator	Free tier / Paid (metered)	Yes	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Adzuna (US)	Free-tier daily cap; ≤ 50/page, no ceiling found	United States (configured)	~2,000/run (self-capped)	Low	Keep; add date/category filters
USAJOBS	Hard 10,000 result window	US federal only	10,000 (at cap)	Low	Keep; partition by category/location
Greenhouse/Lever/Ashby/SmartRecruiters/Workable/Teamtailor	None on most; some real pagination	US-heavy, some global	~130,000+ combined	Low	Keep; grow curated company-slug lists
Jooble/Careerjet/SerpApi/OpenWeb Ninja	Varies; wildcard queries used today	Global (US-weighted)	~1,100-2,000 each	Low	Keep; replace wildcard with targeted queries

United States — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
CareerOneStop / NLx "List Jobs V2"	Government (US DOL, NLx)	Free (partner-gated)	Yes (token; approval needed)	Yes
Fantastic Jobs / Active Jobs DB	Commercial aggregator (ATS-sourced)	Free trial; ~\$1/1,000 jobs	Yes	Yes
Coresignal	Enterprise/Commercial	Paid (from \$49/mo)	Yes	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
CareerOneStop / NLx "List Jobs V2"	Not fully published	US, non-federal, national	Large (authoritative)	Medium	Investigate/Add if approved
Fantastic Jobs / Active Jobs DB	Generous; hourly refresh	US + English-speaking	3M+/month globally	Low	Add if budget allows
Coresignal	Per plan	Global incl. US	399M+ multi-source records	Medium	Enterprise-only; scale play

Canada — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
(none dedicated)	—	—	—	—

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
(none dedicated)	—	Incidental only, via global ATS boards	~3,436 incidental (2.2%)	—	Not applicable — see below

Canada — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
Adzuna 'ca'	Public/Commercial aggregator	Free tier (existing key)	Yes (already held)	Yes
Job Bank (ESDC) feed / Open Gov dataset	Government (Canada)	Free (feed gated; dataset open)	Partner approval (feed); none (dataset)	Yes
ATS company-slug expansion (Canadian-HQ)	ATS	Free (no auth)	No	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Adzuna 'ca'	Same pattern as US Adzuna	Canada	~2,000/run	Trivial	Add — Critical priority
Job Bank (ESDC) feed / Open Gov dataset	Feed unknown; dataset refreshes monthly	Canada (national)	Very large (100,000+)	Medium-High	Investigate — authoritative, freshness-limited
ATS company-slug expansion (Canadian-HQ)	None/per-platform	Canada	Low thousands	Low	Add — complementary

United Kingdom — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
Reed	Public/Commercial (UK board)	Free tier (keyed)	Yes (Basic Auth)	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Reed	~9,900-job window (errors past boundary)	United Kingdom	~9,000 (near ceiling)	Low	Keep; partition by region/keyword

United Kingdom — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
Adzuna 'gb'	Public/Commercial aggregator	Free tier (existing key)	Yes (already held)	Yes
GOV.UK Find a Job (DWP)	Government (UK)	Free (access model unclear)	Unclear	Partial
Careerjet (UK locale)	Commercial aggregator (integrated)	Free tier (existing key)	Yes (already held)	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Adzuna 'gb'	Same pattern as other Adzuna countries	United Kingdom	~2,000/run	Trivial	Add — de-risks single-source dependency on Reed
GOV.UK Find a Job (DWP)	Unknown	United Kingdom (national)	Potentially large	Unclear	Investigate — confirm read-access model
Careerjet (UK locale)	20/page fixed; no ceiling found	UK (via locale_code)	~2,000/run	Low	Add — optional, modest gain

Europe — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
Bundesagentur für Arbeit	Government (Germany)	Free (public key)	Yes (non-secret key)	Yes
Arbeitsnow	Public (EU/Germany-weighted)	Free (no auth)	No	Yes
Greenhouse/Lever/Ashby/Workable (incidental EU)	ATS	Free (no auth)	No	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Bundesagentur für Arbeit	Hard 10,000 result window	Germany	10,000 (at cap)	Low	Keep; partition by region/keyword
Arbeinnow	None found; full board fetched	Germany-weighted , some EU	~930 (full board)	Low	Keep; already exhausted
Greenhouse/Lever/Ashby/Workable (incidental EU)	None/per-platform	EU (incidental)	Meaningful, uncounted precisely	Low	Keep; grow EU-company slugs

Europe — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
Adzuna multi-country (fr/es/nl/it/pl/at)	Public/Commercial aggregator	Free tier (existing key)	Yes (already held)	Yes
France Travail (ex-Pôle emploi)	Government (France)	Free (registration required)	Yes (OAuth2)	Yes
EURES	Government/EU-wide	Free (no supported pull API)	N/A	Partial

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Adzuna multi-country (fr/es/nl/it/pl/at)	Same free-tier pattern per country	France, Spain, Netherlands, Italy, Poland, Austria	~2,000/run per country	Trivial	Add — Critical, top European lever
France Travail (ex-Pôle emploi)	Not fully published	France	Very large (500,000+)	Medium	Add — authoritative French coverage
EURES	N/A	EU-wide (~3M jobs)	Very large if accessible	High	Not recommended — scraping-only today

India — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
(none dedicated)	—	—	—	—

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
(none dedicated)	—	Incidental only, via global ATS + wildcard aggregators	~3,471 incidental (2.2%)	—	Not applicable — see below

India — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
ATS company-slug expansion (Indian-HQ)	ATS	Free (no auth)	No	Yes
India-scoped query on JSearch / SerpApi	Commercial aggregator (integrated)	Free tier / Paid (metered, existing keys)	Yes (already held)	Yes
National Career Service (data.gov.in)	Government (India)	Free (open dataset)	None (dataset)	Yes
Fantastic Jobs / Active Jobs DB	Commercial aggregator	Free trial; ~\$1/1,000 jobs	Yes	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
ATS company-slug expansion (Indian-HQ)	None/per-platform	India	Low thousands	Low	Add — Critical, best free route
India-scoped query on JSearch / SerpApi	200 req/mo (JSearch); 250 searches/mo (SerpApi)	India (via country filter)	Modest, budget-bound	Low	Add — cheapest India-specific volume
National Career Service (data.gov.in)	N/A — periodic bulk dataset	India (national)	Large in aggregate	Medium-High	Investigate — authoritative, freshness-limited
Fantastic Jobs / Active Jobs DB	Generous; hourly refresh	Global incl. India	Large (part of 3M+/mo global)	Low	Add if budget allows

Singapore / Southeast Asia — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
(none dedicated)	—	—	—	—

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
(none dedicated)	—	Incidental only, via global ATS + wildcard aggregators	~4,629 incidental (3.0%)	—	Not applicable — see below

Singapore / Southeast Asia — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
MyCareersFuture	Government (Singapore, GovTech)	Free (no auth)	No	Community-documented
ATS company-slug expansion (SEA-HQ)	ATS	Free (no auth)	No	Yes
Adzuna 'sg'	Public/Commercial aggregator	Free tier (existing key, if covered)	Yes (already held)	Yes
JobStreet / JobsDB (SEEK group)	Commercial (dominant SEA boards)	No public API (partner only)	N/A	N/A

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
MyCareersFuture	Unknown/none published; 100/page	Singapore	~80,000+ available	Low	Add — Critical, top SEA priority
ATS company-slug expansion (SEA-HQ)	None/per-platform	Singapore + broader SEA	Low thousands	Low	Add — complementary
Adzuna 'sg'	Same pattern as other Adzuna countries	Singapore	~2,000/run if available	Trivial	Add if available
JobStreet / JobsDB (SEEK group)	N/A	MY, ID, PH, SG, TH, VN	Very large if accessible	High	Not recommended — scraping-only

LATAM — Existing APIs already implemented

API Name	Type	Pricing	Auth	Docs
(none dedicated)	—	—	—	—

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
(none dedicated)	—	Incidental only, mostly remote/nearshore	~6,955 incidental (4.5%)	—	Not applicable — see below

LATAM — Additional APIs that could improve coverage

API Name	Type	Pricing	Auth	Docs
Adzuna 'br' + 'mx'	Public/Commercial aggregator	Free tier (existing key)	Yes (already held)	Yes
Get on Board	Commercial (LATAM tech board)	Likely free/low-cost (unverified)	Likely yes	Advertised, not independently verified
Himalayas search endpoint (country filter)	Public (integrated, richer endpoint unused)	Free (no auth)	No	Yes
ATS company-slug expansion (LATAM-HQ)	ATS	Free (no auth)	No	Yes

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Adzuna 'br' + 'mx'	Same pattern as other Adzuna countries	Brazil, Mexico	~2,000/run per country	Trivial	Add — Critical, top LATAM lever
Get on Board	Unknown	LATAM (tech-focused)	Moderate (est. 5,000-20,000)	Low-Medium	Investigate/Add — one LATAM board with a confirmed real API

API Name	Rate Limits / Pagination	Country Coverage	Expected Volume	Effort	Recommendation
Himalayas search endpoint (country filter)	20/page fixed; 90,000+ archive	Remote-LATAM (via country filter)	Subset of 90,000+ archive	Low	Add — cheap, reuses existing integration
ATS company-slug expansion (LATAM-HQ)	None/per-platform	LATAM	Low thousands	Low	Add — complementary

17. Existing Provider Optimization

Every provider, analyzed for current volume, realistic ceiling, and the mechanism to collect more. Figures match research/api_analysis.md §20 and Section 13 exactly.

Provider	Current Jobs	Estimated Maximum	Additional Jobs Possible	Difficulty	Recommended Strategy
Arbeitsnow	930	~950-1,000	~0-100	N/A	None — full board already exhausted; no filter dimension exists
Himalayas	3,000	13,000-23,000	+10,000-20,000	Low	Switch to /search endpoint; partition by country/category
RemoteOK	100	~100	~0	N/A	None confirmed; verify tag-scoped endpoints before assuming more
Jobicy	100	300-500	+200-400	Low	Partition by geo/industry/tag filters across multiple queries
The Muse	2,000	5,000-8,000	+3,000-6,000	Medium	Partition by category/location filters; consider an API key
Bundesagentur	10,000	25,000-40,000	+15,000-30,000	Medium	Partition by region (radius/location) or keyword across multiple queries
Jooble	1,129	4,000-9,000	+3,000-8,000	Low	Replace wildcard with targeted keyword and location searches
USAJOBS	10,000	20,000-30,000	+10,000-20,000	Medium	Partition by category or location filters across multiple queries
Adzuna	2,000	22,000-42,000	+20,000-40,000	Low	Query multiple country endpoints already covered by the existing key
Reed	9,000	17,000-24,000	+8,000-15,000	Medium	Partition by UK region (location) or keyword across multiple queries
SerpApi	50	~2,500/mo (budget-bound)	Budget-dependent	Low	Reallocate quota to targeted location/date-filtered queries
OpenWeb Ninja	~47	~200/mo (budget-bound)	Budget-dependent	Low	Reallocate quota toward country-filtered (e.g. India) queries
Careerjet	2,000	7,000-12,000	+5,000-10,000	Low	Add locale/country-scoped passes plus real keyword searches
TheirStack	0	0 (plan-gated)	0 until plan restored	N/A	Resolve the account/billing restriction first
Greenhouse	22,728	27,728-37,728	+5,000-15,000	Low	ATS company expansion — add curated slugs (no query filters exist)
Lever	17,755	22,755-27,755	+5,000-10,000	Low	ATS company expansion — add curated company slugs
Ashby	5,171	7,171-10,171	+2,000-5,000	Low	ATS company expansion — source slugs outside AI/SaaS/FinTech
SmartRecruiters	65,193	70,193-80,193	+5,000-15,000	Low	ATS company expansion — add slugs, weigh against concentration risk
Workable	7,644	10,644-14,644	+3,000-7,000	Low	ATS company expansion — source slugs in staffing/consulting/fintech/healthcare
Teamtailor	474	974-1,974	+500-1,500	Low	ATS company expansion — source additional career sites (highest candidate hit rate)

How additional jobs can be collected, mapped to technique

- Multiple country endpoints: Adzuna (10+ countries already covered by the existing key), Careerjet (locale_code for UK/India/Brazil/SEA locales).
- Category searches: The Muse (category), USAJOBS (JobCategoryCode), Himalayas (/search endpoint).
- Company searches: TheirStack (company filter, once plan-restored); all six ATS providers via company-slug expansion (their only lever, since none expose query filters).
- Location filters: Adzuna (where), Bundesagentur (wo), USAJOBS (LocationName), Reed (locationName), Himalayas (/search).

- Keyword searches: Jooble, Careerjet, Adzuna (what), Bundesagentur (was), USAJOBS (Keyword) — replacing today's wildcard/unscoped queries with real terms.
- Date filters: Adzuna (max_days_old), USAJOBS (DatePosted), Bundesagentur (veroeffentlichzeit), Jooble (datecreatedfrom), TheirStack (posted_at_max_age_days) — a freshness lever more than a volume one.
- Remote filters: USAJOBS (RemoteIndicator), JSearch/OpenWeb Ninja (remote_jobs_only), SerpApi (work_from_home) — a quality lever, not a volume one.
- Department filters: not offered as a request-side filter by any provider (Ashby/SmartRecruiters/Workable return it only as data), so it can't expand volume.
- Radius searches: Bundesagentur (umkreis), Reed (distance), USAJOBS (Radius).
- Multiple API queries (partitioning): the core mechanism for Adzuna, Bundesagentur, USAJOBS, Reed, The Muse, Himalayas, Jooble, Careerjet — each hits a hard per-query ceiling only partitioning can exceed.
- Incremental synchronization: improves freshness rather than raw volume; most relevant where a genuine updated/date field exists (Jooble updated, Greenhouse updated_at, USAJOBS if ApplicationCloseDate is extracted).
- ATS company expansion: the only lever for Greenhouse, Lever, Ashby, SmartRecruiters, Workable, and Teamtailor — none of the six expose any query filter, so growing their curated company-slug lists is the sole path to more volume from these six providers.

18. Regional Coverage Improvement Plan

Current state and gaps

Region	Current Coverage	Coverage Gaps	Best Existing Provider to Optimize
United States	Strong (51.0%; deepest ATS + federal coverage)	Broad but curated-company-driven, not query-driven	Adzuna (US) — add date/category filters
Canada	Weakest target region (2.2%, incidental only)	No dedicated source at all	Adzuna — enable 'ca' country
United Kingdom	Well covered for its size (6.6-8.2%) but single-source	No resilience/backup source	Reed — partition by region/keyword
Europe	14.0-18.7%, but 56% Germany-concentrated	Rest of Europe (FR/ES/NL/IT/PL) thin	Bundesagentur — partition by region
India	2.2% incidental; 4th-largest country despite no dedicated source	No dedicated source; consumer boards have no public API	ATS providers — expand Indian-HQ slugs
Singapore / Southeast Asia	3.0%, weakest alongside Canada	No dedicated source anywhere in SEA	ATS slug expansion (no dedicated provider to optimize)
LATAM	4.5%, mostly remote/nearshore, not in-country	Real in-country volume (Brazil/Mexico) thin	Himalayas — use /search with country filter

Recommended action and priority

Region	Best New API to Integrate	Expected Increase	Eng. Effort	Business Value	Priority
United States	CareerOneStop/NLx (authoritative non-federal)	+10,000-20,000 (if approved)	Medium	Medium (already strong)	Medium
Canada	Job Bank Canada (ESDC)	+15,000-25,000 combined	Trivial	Very High	Critical
United Kingdom	Adzuna 'gb'	+10,000-17,000	Low-Medium	High	High
Europe	Adzuna multi-country + France Travail	+25,000-42,000 combined	Low-Medium	High	Critical
India	India-scoped JSearch query (cheapest lever)	+10,000-20,000	Low	High	High
Singapore / Southeast Asia	MyCareersFuture (Singapore)	+15,000-20,000	Low	Very High	Critical
LATAM	Adzuna 'br' + 'mx'	+10,000-15,000	Trivial-Low	High	High

19. ROI Prioritization

All recommendations — existing-provider optimization and new-API integration — ranked into five priority tiers.

Priority	Recommendation Bundle	Est. Additional Jobs	Eng. Effort	Maintenance Effort	Business Impact	Technical Complexity
P1	Adzuna multi-country expansion (US-only -> CA/GB/DE/FR/ES/NL/IT/PL/AT/BR/MX)	+20,000-40,000	Low	Low	Very High	Low
P2	MyCareersFuture (Singapore) + ATS company-slug expansion (Canada/India/SEA/LATAM)	+25,000-45,000 combined	Low-Medium	Medium	Very High	Low
P3	Query partitioning on already-capped providers (Bundesagentur, USAJOBS, Reed, Himalayas, Jooble, Careerjet, The Muse)	+56,000-99,000 combined	Medium	Low-Medium	High	Medium
P4	Freshness/incremental-sync engineering (run-to-run diffing, date filters, scheduling)	+0 net-new (fixes 46% stale problem)	Medium	Medium	Very High (data quality)	Medium
P5	New official/government APIs requiring approval (CareerOneStop/NLx, France Travail, Job Bank Canada feed, National Career Service India)	+20,000-40,000 if all approved	Medium-High	Medium	Medium-High	Medium-High

Management Summary

- Which existing APIs first? Adzuna (multi-country) is the single highest-leverage change, followed by Bundesagentur, USAJOBS, Reed, Himalayas, Jooble, and Careerjet — all capped with unused filters.
- Which APIs are already at their limit? Arbeitnow and RemoteOK (no query-based lever left); TheirStack is at zero due to billing, not an API ceiling.
- Which new APIs first? MyCareersFuture (Singapore, no-auth, free, ~15,000+) first, then Job Bank Canada and France Travail (authoritative, but need partner/OAuth approval).
- Which regions need attention? Canada and Singapore/SEA (~2-3%, zero dedicated source), then India (4th-largest country, no dedicated source) and LATAM (mostly remote/nearshore, not real in-country volume).
- How many more jobs before new providers? ~75,000-140,000, purely from the 8 already-integrated providers' untapped filter/partitioning capacity — consistent across Section 13 and research/api_analysis.md §§14,19,20.

20. Roadmap

Recommendations are grouped into five priorities, each with estimated coverage increase, engineering effort, and business value.

Priority	Focus	Net-new jobs	Effort	Business value
P1	Adzuna multi-country + date filter	15,000–25,000	Low	Very High
P2	Freshness foundation (scheduling + expiry)	0 (quality fix)	Medium	Very High
P3	Weak regions: MyCareersFuture + ATS slugs + India query	40,000–70,000	Low–Medium	High
P4	Result-window bypass via partitioning + France Travail	20,000–50,000	Medium	High
P5	Dedup/location normalization + optional paid aggregator	Quality (+50,000 optional)	Medium	Medium–High

If only two weeks of engineering time were available

- Days 1–2: Ship Adzuna multi-country + max_days_old (Priority 1).
- Days 2–3: Capture a fetched_at timestamp and surface a “data as of” signal.
- Days 3–6: Build the scheduling + presence-diff expiry core (Priority 2).
- Days 6–9: Add MyCareersFuture (Singapore) and a first batch of Indian/SEA/Canada ATS slugs (Priority 3).
- Days 9–12: Prove out one result-window partition (e.g., Reed or The Muse) plus URL normalization in dedup.
- Days 12–14: Integration testing, dedup verification, and rate-limit/budget tuning across the newly added passes.

This sequence realistically adds ~40,000–70,000 net-new, better-distributed jobs and fixes freshness within two weeks — no scraping, no new commercial contracts.

21. Future Improvements

Beyond the two-week roadmap, structural investments that compound value as the dataset grows:

- A persistent datastore (e.g., Postgres or OpenSearch) replacing the single jobs.json file, paired with a paginated, queryable search API — the current design of loading the entire file into the browser will not scale past the low hundreds of thousands of jobs.
- Incremental/delta ingestion with per-job first_seen/last_seen tracking and automated expiry, replacing the current full-overwrite snapshot model.
- Scalable deduplication — normalized-URL keys plus blocked fuzzy/locality-sensitive-hash matching — to handle cross-provider overlap at higher volume, and description-based matching once job descriptions are added to the schema.
- Automated ATS company/slug discovery and health monitoring, so coverage grows without linear manual curation effort.
- A decoupled, per-source pipeline with queueing and rate-limit budgeting, so one slow or large provider cannot stall an entire run, with observability on volume, freshness, and duplicate rate per source per run.
- Optional adoption of a paid aggregator (e.g., Fantastic Jobs' Active Jobs DB) for a step-change in both volume and freshness once budget allows.

22. Recent Job Market Analysis (Last 30 Days)

A second, independently-computed pass over the same <=30-day slice as Sections 7 and 14 (84,077 jobs, 54.0% of the dataset), included for cross-validation. Methodology and reasoning are as in Section 14; only the resulting figures are restated below.

1. Job Category Distribution

Same category methodology as Section 14.

Category	Recent Jobs	% of recent
Other	44,858	53.4%
Supply Chain	7,531	9.0%
Software Engineering	5,571	6.6%
Customer Support	5,264	6.3%
Healthcare	3,505	4.2%
Sales	2,888	3.4%
Executive	1,570	1.9%
Education	1,474	1.8%
Accounting	1,224	1.5%
Project Management	994	1.2%
Internship / Graduate	911	1.1%
Manufacturing	792	0.9%
Human Resources	779	0.9%
Marketing	755	0.9%
Product Management	576	0.7%
Finance	525	0.6%
DevOps / Cloud	517	0.6%
Data Science / Analytics	511	0.6%
Legal	399	0.5%
Business Development	392	0.5%
Cybersecurity	374	0.4%
Administrative	334	0.4%

Category	Recent Jobs	% of recent
Customer Success	302	0.4%
AI / Machine Learning	299	0.4%
Digital Marketing	293	0.3%
Operations	274	0.3%
Procurement	272	0.3%
Recruiting	250	0.3%
IT Support	219	0.3%
UI / UX Design	174	0.2%
Government	102	0.1%
Pharmaceutical	77	0.1%
Graphic Design	71	0.1%

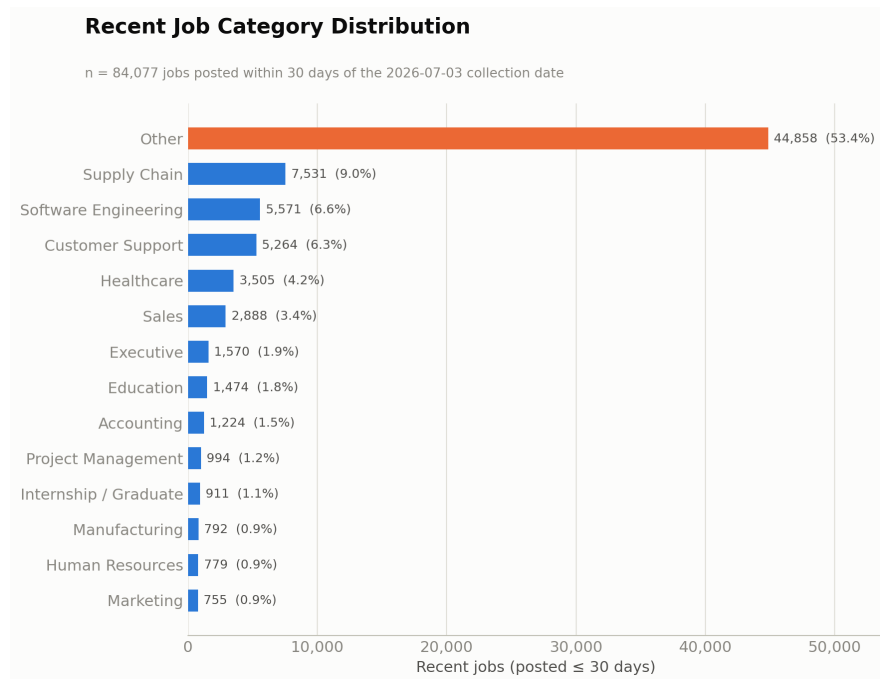


Figure 9. Recent job category distribution (posted within 30 days).

This pass: Supply Chain (9.0%), Software Engineering (6.6%), Customer Support (6.3%), Healthcare (4.2%), Sales (3.4%) — consistent with Section 14's ranking within normal classifier variance.

2. Recent Jobs by Region

Region	Recent Jobs	% of recent
United States	39,124	46.5%
Europe	15,721	18.7%
Other	14,372	17.1%
United Kingdom	6,925	8.2%
LATAM	2,886	3.4%
Singapore / Southeast Asia	1,744	2.1%
India	1,656	2.0%
Canada	1,649	2.0%

Recent Jobs by Region

n = 84,077 jobs posted within 30 days

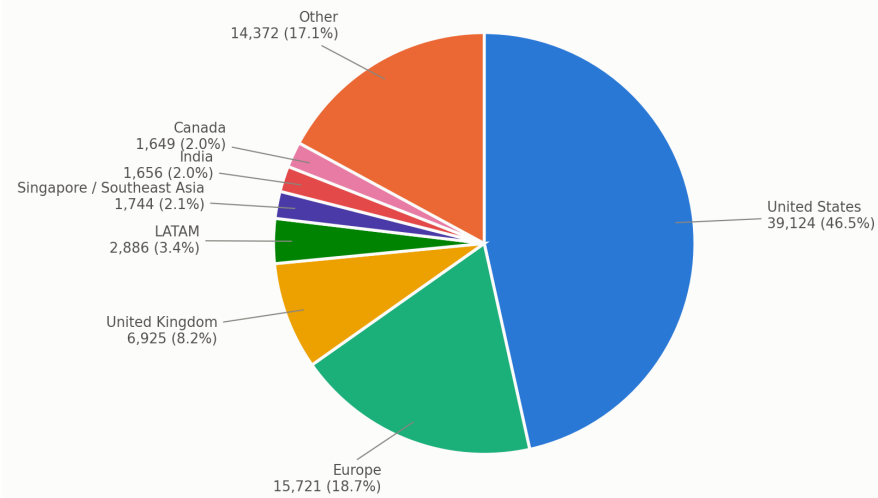


Figure 10. Recent jobs by region (posted within 30 days).

Same regional pattern as Section 14: US 46.5%, Europe 18.7% (Germany-heavy), UK 8.2% (Reed); Canada/India/SEA each ~2%.

3. Recent Jobs by Provider

Provider	Recent Jobs	% of recent	Freshness rate (of provider's own total)
SmartRecruiters	36,479	43.4%	56.0%
Reed	8,658	10.3%	96.2%
Bundesagentur	7,800	9.3%	100.0%
USAJOBS	6,138	7.3%	61.4%
Other (SerpApi, OpenWeb Ninja, CareerJet, TheirStack, misc.)	5,091	6.1%	50.4%
Greenhouse	5,013	6.0%	33.6%
Jooble	2,973	3.5%	95.1%
Himalayas	2,964	3.5%	100.0%
Lever	2,653	3.2%	14.9%
Ashby	1,688	2.0%	32.6%
Adzuna	1,467	1.7%	73.3%
Workable	1,232	1.5%	29.6%
Arbeitnow	930	1.1%	100.0%
The Muse	615	0.7%	30.8%
Teamtailor	176	0.2%	38.6%
Jobicy	100	0.1%	100.0%
RemoteOK	100	0.1%	100.0%

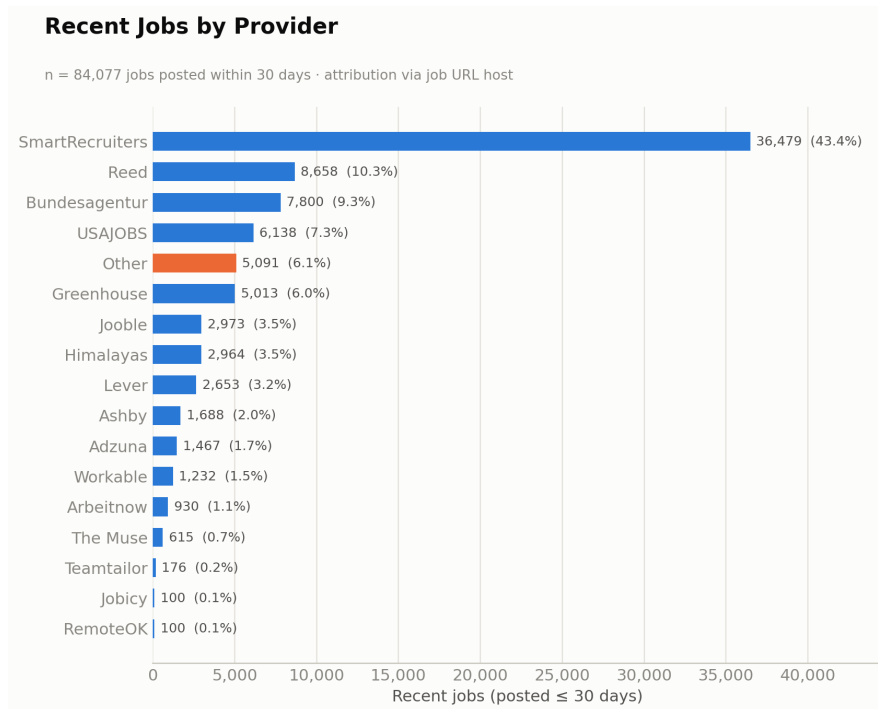


Figure 11. Recent jobs by provider (posted within 30 days).

By volume: SmartRecruiters (43.4%), Reed (10.3%), Bundesagentur (9.3%). By freshness rate (more diagnostic): same five providers as Section 7/14 are ~100% fresh; Lever through Greenhouse remain lowest. See Section 7 for the full explanation.

4. Recent Jobs by Company

Rank	Company	Recent Jobs	% of recent
1	Domino's	20,620	24.5%
2	AccorHotel	3,397	4.0%
3	AECOM	2,297	2.7%
4	Bosch Group	1,900	2.3%
5	Veterans Health Administration	1,675	2.0%
6	Anduril Industries	817	1.0%
7	Northwestern Memorial Healthcare	751	0.9%
8	Reed	689	0.8%
9	SpaceX	681	0.8%
10	Delivery Hero	599	0.7%
11	eFinancialCareers	514	0.6%
12	Primark	454	0.5%
13	ASSYSTEM	417	0.5%
14	Tesco	396	0.5%
15	AUREA GmbH	341	0.4%
16	Boyd Gaming	335	0.4%
17	Munson Healthcare	328	0.4%
18	NielsenIQ	324	0.4%
19	Hays Specialist Recruitment Limited	320	0.4%
20	Continental	309	0.4%

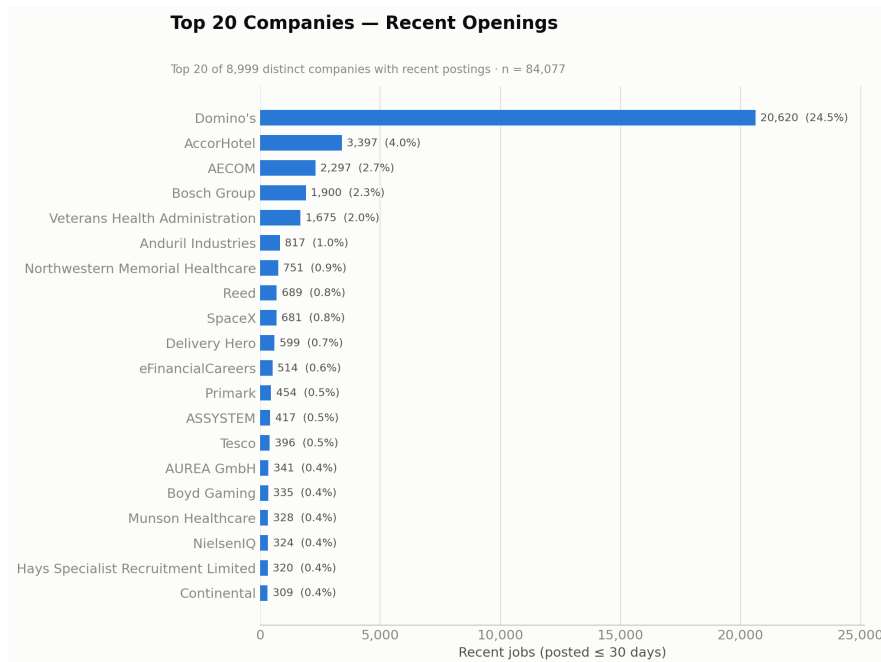


Figure 12. Top 20 companies by recent openings.

Matches Section 14: top 10 companies are 39.8% of recent postings, Domino's alone 24.5% (vs. 15.7% dataset-wide), among 8,999 distinct recent-posting companies.

5. Fresh Job Recommendations

- Freshest APIs: Bundesagentur, Himalayas, Arbeitnow, RemoteOK, Jobicy (~100%), then Reed (96.2%) and Jooble (95.1%) — see Section 7 for why.
- Mostly older jobs: Lever, Workable, The Muse, Ashby, Greenhouse (14.9–33.6% fresh) — full-board ATS fetches (Section 7).
- Those five benefit least from more frequent runs alone — incremental sync, not a tighter schedule, is the fix.
- Sync daily: SmartRecruiters, Reed, Bundesagentur, USAJOBS, Jooble.
- Incremental-sync candidates with an existing but unused date filter: USAJOBS, Adzuna, Jooble, Bundesagentur, TheirStack.

6. Engineering Recommendations

- Date filtering: enable existing filters on USAJOBS, Adzuna, Bundesagentur, Jooble, TheirStack (see Section 12).
- Incremental sync: the single biggest lever for Lever, Workable, The Muse, Ashby, Greenhouse (full detail in Section 7 §5).
- Region splitting: partition Adzuna, Bundesagentur, USAJOBS, and Reed by country/region (Section 12).
- Keyword searches: replace wildcard queries (Jooble, Careerjet, SerpApi, OpenWeb Ninja) with targeted keywords.
- Company-based searches: for the six ATS providers, the only lever is slug curation plus run-over-run diffing.
- Category searches: partition The Muse, USAJOBS, and Himalayas by category alongside date ordering.

Net estimate: these fixes target ~46,000 jobs that are stale despite coming from providers capable of freshness, realistically raising the overall ≤30-day-fresh share from 54.0% into the high-60s–low-70s% range without adding a new provider.

23. Conclusion

The platform is a mature, well-engineered first version: a clean, extensible 20-provider architecture behind one shared interface, a fully normalized schema, correct URL-based deduplication, and disciplined per-source error isolation. It is production-ready as a periodic bulk job board, but not yet freshness- or scale-sensitive — a static snapshot with no scheduling, expiry, or persistent per-job state.

Regional strength concentrates in the US, Germany, and (for its size) the UK; Canada and SEA are thin, India has no dedicated source, and the UK rides on a single provider. Duplicate handling is excellent at the URL level. Freshness is the weakest dimension and highest-value fix — an architectural gap, not a sourcing one.

Nearly all high-impact opportunities identified are already unlocked: multi-country Adzuna, MyCareersFuture, France Travail, ATS slug expansion, and India coverage via existing infrastructure — all free/low-cost, legally clean, no new commercial relationships.

On scalability: the architecture cannot scale to millions of jobs as built (54 MB at 156,000 jobs implies ~350 MB at 1M, >1 GB at 3M — both storage and the frontend's whole-file load need to change). It can comfortably grow into the low hundreds of thousands with this roadmap, and the modular source design makes a future datastore-backed, incrementally synced service a natural evolution, not a rewrite.

24. Provider Capability Matrix

This section provides a management-friendly comparison of every implemented provider, computed directly from the 155,794 job records currently in jobs.json. Provider attribution uses a rigorous method that goes beyond a simple URL-host lookup: each job's URL host is checked first, and for the remainder, the company field is cross-checked against each ATS source module's own curated company-slug list (sources/greenhouse.py, lever.py, ashby.py, workable.py, teamtailor.py, smartrecruiters.py) - because several providers' APIs return the employer's own domain rather than the platform's (e.g. Greenhouse's absolute_url frequently points to stripe.com, not boards.greenhouse.io). This recovers 6,780 Greenhouse jobs and 2,191 Bundesagentur jobs (the latter via Bundesagentur's documented externeUrl redirect to the employer's own German job-board listing) that a naive host count would have misattributed to "Other." Only 1,125 jobs (0.7% of the dataset) remain genuinely unattributable to a specific provider - these are most likely SerpApi, OpenWeb Ninja, and Careerjet results, which return the original posting's own URL rather than an aggregator domain, and cannot be separated further by URL alone. Region and job-category are computed by a location/title keyword classifier built specifically for this section; totals differ modestly from the region/category figures used elsewhere in this report because of classifier differences, not because of different underlying data. No Python source file and no jobs.json regeneration were involved in producing this section.

Part 1. Provider Overview

Every registered provider, with total jobs, its dominant regions/categories, remote and freshness rates, and authentication/API-type facts (drawn from research/api_analysis.md).

Provider	Total Jobs	% of Dataset	Primary Regions	Primary Categories	Remote %	Recent % (≤30d)	Auth Required	API Type
Smart Recruiters	65,193	41.8%	United States, Other / Unclassified	Other, Operations	3.0%	56.0%	No (public posting endpoint)	ATS
Greenhouse	21,722	13.9%	United States, Other / Unclassified	Other, Tech	16.3%	33.7%	No	ATS
Lever	17,755	11.4%	United States, Other / Unclassified	Other, Sales	19.8%	14.9%	No (GET)	ATS
USAJOBS	10,000	6.4%	United States	Other, Healthcare	0.2%	61.4%	Yes (3 headers)	Government
Bundesagentur	9,991	6.4%	Germany, Europe (Other)	Other, Operations	0.3%	100.0%	Yes (public, non-secret key)	Government
Reed	9,000	5.8%	United Kingdom	Other, Operations	0.2%	96.2%	Yes (HTTP Basic)	Commercial
Ashby	5,171	3.3%	United States, Other / Unclassified	Other, Tech	59.6%	32.6%	No	ATS

Provider	Total Jobs	% of Dataset	Primary Regions	Primary Categories	Remote %	Recent % ($\leq 30d$)	Auth Required	API Type
Workable	4,159	2.7%	United States, Singapore / SEA	Other, Tech	28.5%	29.6%	No (public widget API)	ATS
Jooble	3,127	2.0%	United Kingdom, United States	Other, Operations	0.2%	95.1%	Yes (key in URL)	Commercial aggregator
Himalayas	2,964	1.9%	United States, Other / Unclassified	Other, Tech	100.0%	100.0%	No	Public
Adzuna	2,001	1.3%	United States, Other / Unclassified	Healthcare, Operations	0.8%	73.3%	Yes (app_id + app_key)	Commercial aggregator
The Muse	2,000	1.3%	United States, Other / Unclassified	Other, Healthcare	0.5%	30.8%	No (key optional)	Public/Commercial aggregator
Arbeithnow	930	0.6%	Germany, Other / Unclassified	Other, Tech	14.3%	100.0%	No	Public
Teamtailor	456	0.3%	Other / Unclassified, Europe (Other)	Other, Operations	24.8%	38.6%	No (public RSS)	ATS
RemoteOK	100	0.1%	Other / Unclassified, United States	Other, Tech	100.0%	100.0%	No (requires browser User-Agent)	Public
Jobicy	100	0.1%	United States, Other / Unclassified	Other, Tech	100.0%	100.0%	No	Public
TheirStack	0	0.0%	n/a (0 jobs)	n/a (0 jobs)	n/a	n/a	Yes (Bearer token)	Commercial (metered)
Unattributed	1,125	0.7%	United States, Other / Unclassified	Other, Tech	16.8%	52.2%	Yes (mixed - SerpApi/OpenWeb Ninja/Careerjet)	Commercial aggregator (metered: SerpApi/OpenWeb Ninja/Careerjet)

Total: 155,794 jobs across 17 attributable buckets - 16 named providers, 1,125 unattributable jobs, and TheirStack (0 jobs, plan-gated with HTTP 402). SmartRecruiters alone supplies 41.8% of the dataset; the top three providers (SmartRecruiters, Greenhouse, Lever) together supply 67.1%.

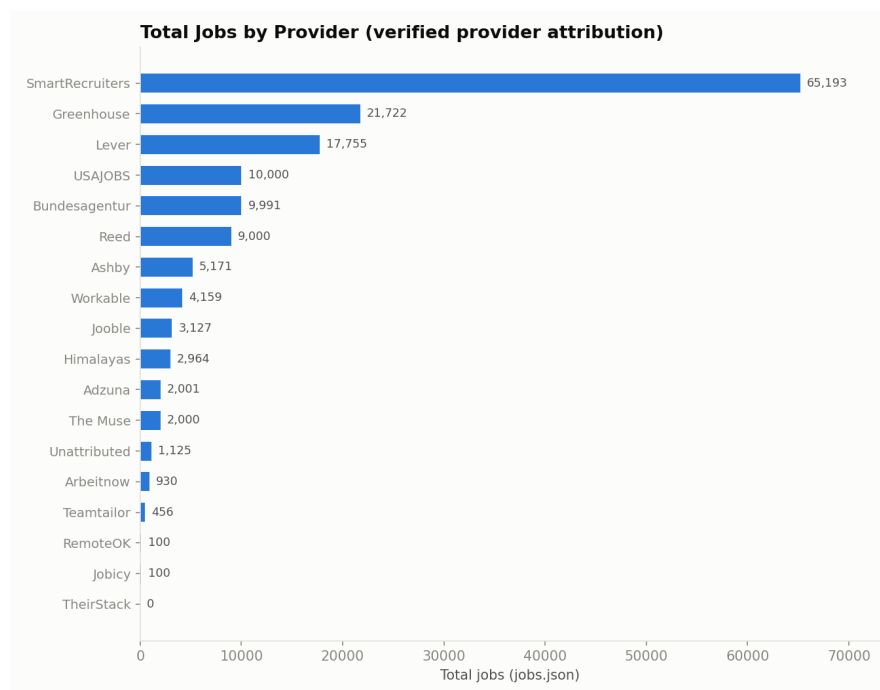


Figure 17. Total jobs by provider (verified attribution).

Part 2. Provider vs. Region Matrix

Which regions each provider primarily contributes to, as a percentage of that provider's own jobs (row-normalized). Levels: High $\geq 40\%$, Medium 15-39%, Low 5-14%, Minimal 1-4%, None $< 1\%$.

Provider	United States	Canada	United Kingdom	Germany	Europe (Other)	India	Singapore / SEA	LATAM	Remote / Global	Other / Unclassified
SmartRecruiters	High (63%)	Minimal (2%)	Low (6%)	Minimal (2%)	Low (7%)	Minimal (4%)	Minimal (5%)	Minimal (2%)	None (0%)	Low (10%)
Greenhouse	High (60%)	Minimal (4%)	Minimal (4%)	Minimal (1%)	Low (5%)	Minimal (4%)	Minimal (2%)	Minimal (2%)	Low (6%)	Low (11%)
Lever	High (63%)	Minimal (3%)	Minimal (2%)	Minimal (2%)	Minimal (4%)	Minimal (2%)	Minimal (2%)	Minimal (1%)	Minimal (2%)	Medium (20%)
USAJOBS	High (100%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)
Bundesagentur	None (0%)	None (0%)	None (0%)	High (93%)	Low (7%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)
Reed	None (0%)	None (0%)	High (100%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)
Ashby	Medium (32%)	Low (5%)	Low (8%)	Minimal (1%)	Low (6%)	Minimal (2%)	Minimal (5%)	None (1%)	Low (9%)	Medium (31%)
Workable	High (54%)	Low (7%)	Minimal (3%)	Minimal (1%)	Minimal (5%)	Minimal (3%)	Low (13%)	Minimal (3%)	None (0%)	Low (11%)
Jooble	Medium (35%)	None (0%)	High (43%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	Medium (21%)
Himalayas	High (54%)	Minimal (4%)	Low (6%)	Low (5%)	Low (8%)	Minimal (4%)	Minimal (4%)	Low (6%)	None (1%)	Low (9%)
Adzuna	High (92%)	None (0%)	Minimal (2%)	None (0%)	None (1%)	None (0%)	None (0%)	None (0%)	None (0%)	Minimal (5%)
The Muse	High (90%)	None (1%)	Minimal (2%)	None (0%)	Minimal (1%)	Minimal (1%)	None (0%)	None (1%)	Minimal (2%)	Minimal (2%)
Arbeitsnow	None (0%)	None (0%)	None (0%)	High (59%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	High (41%)
Teamtailor	Low (12%)	Minimal (1%)	Minimal (5%)	Minimal (3%)	Medium (29%)	None (0%)	Low (10%)	Low (8%)	None (0%)	Medium (32%)
RemoteOK	Medium (28%)	Minimal (4%)	Low (6%)	None (0%)	Low (5%)	Minimal (1%)	Minimal (1%)	Minimal (1%)	Minimal (4%)	High (50%)
Jobicy	High (66%)	Minimal (4%)	None (0%)	Minimal (3%)	Low (10%)	None (0%)	Minimal (4%)	Minimal (1%)	None (0%)	Low (12%)

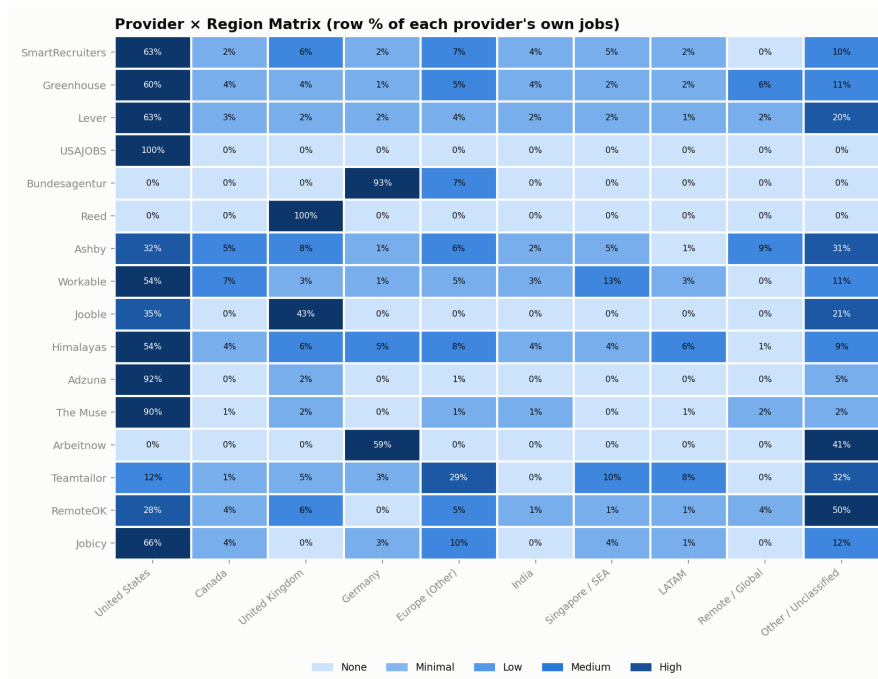


Figure 18. Provider x region heatmap (row % of each provider's own jobs).

Which providers dominate each region: the United States is carried by SmartRecruiters, Greenhouse, and Lever (85,438 jobs combined across the three); Germany is almost exclusively Bundesagentur (9,988 of 12,312 Germany-classified jobs); the United Kingdom is dominated by Reed (9,000 of 16,152) with Joooble a distant second; Canada, India, Singapore/SEA, and LATAM have no provider reaching "Medium" - every one of those four regions is served only incidentally by providers whose primary focus is elsewhere.

Part 3. Provider vs. Job Category Matrix

Which business categories each provider primarily contributes to, as a percentage of that provider's own jobs (row-normalized). Same High/Medium/Low/Minimal/None thresholds as Part 2.

Provider	Tech	Sales	Design	Healthcare	Government	Finance	HR	Operations	Other
SmartRecruiters	Minimal (4%)	Minimal (2%)	None (0%)	Minimal (3%)	Minimal (2%)	Minimal (1%)	Minimal (1%)	Medium (28%)	High (56%)
Greenhouse	Medium (24%)	Low (9%)	None (1%)	None (1%)	None (0%)	Minimal (4%)	Minimal (2%)	Low (8%)	High (48%)
Lever	Low (7%)	Low (13%)	None (0%)	Minimal (4%)	None (0%)	Minimal (2%)	None (1%)	Low (7%)	High (63%)
USAJOBS	Minimal (1%)	None (0%)	None (0%)	Medium (21%)	None (0%)	None (1%)	Minimal (1%)	Minimal (3%)	High (68%)
Bundesagentur	None (1%)	None (1%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	None (1%)	High (97%)
Reed	Minimal (2%)	Minimal (4%)	None (0%)	Minimal (1%)	None (0%)	Low (5%)	None (1%)	Low (6%)	High (74%)
Ashby	Medium (24%)	Low (10%)	Minimal (1%)	None (0%)	None (0%)	Minimal (5%)	Minimal (2%)	Low (8%)	High (47%)
Workable	Low (11%)	Low (8%)	None (1%)	Minimal (4%)	None (0%)	Minimal (3%)	Minimal (2%)	Low (9%)	High (56%)
Joooble	None (0%)	Minimal (1%)	None (0%)	Minimal (3%)	None (0%)	None (0%)	None (0%)	Minimal (4%)	High (90%)
Himalayas	Medium (16%)	Low (9%)	None (1%)	Low (12%)	None (1%)	Minimal (4%)	Minimal (3%)	Low (11%)	Medium (38%)
Adzuna	None (0%)	None (0%)	None (0%)	High (69%)	None (0%)	None (0%)	None (0%)	Medium (16%)	Low (13%)
The Muse	Low (12%)	Low (7%)	None (0%)	Low (13%)	None (0%)	Minimal (1%)	Low (6%)	Low (5%)	High (54%)
Arbeitanow	Low (11%)	Minimal (4%)	None (1%)	None (0%)	None (0%)	Minimal (3%)	Minimal (1%)	Minimal (4%)	High (69%)
Teamtailor	Low (11%)	Low (7%)	None (0%)	Low (7%)	None (0%)	Low (5%)	Minimal (2%)	Medium (15%)	High (50%)
RemoteOK	Medium (22%)	None (0%)	None (0%)	None (0%)	None (0%)	None (0%)	Medium (17%)	Medium (18%)	Medium (35%)
Jobicy	Medium (23%)	Low (13%)	None (0%)	Low (5%)	None (0%)	Minimal (4%)	Minimal (2%)	Medium (16%)	Medium (30%)

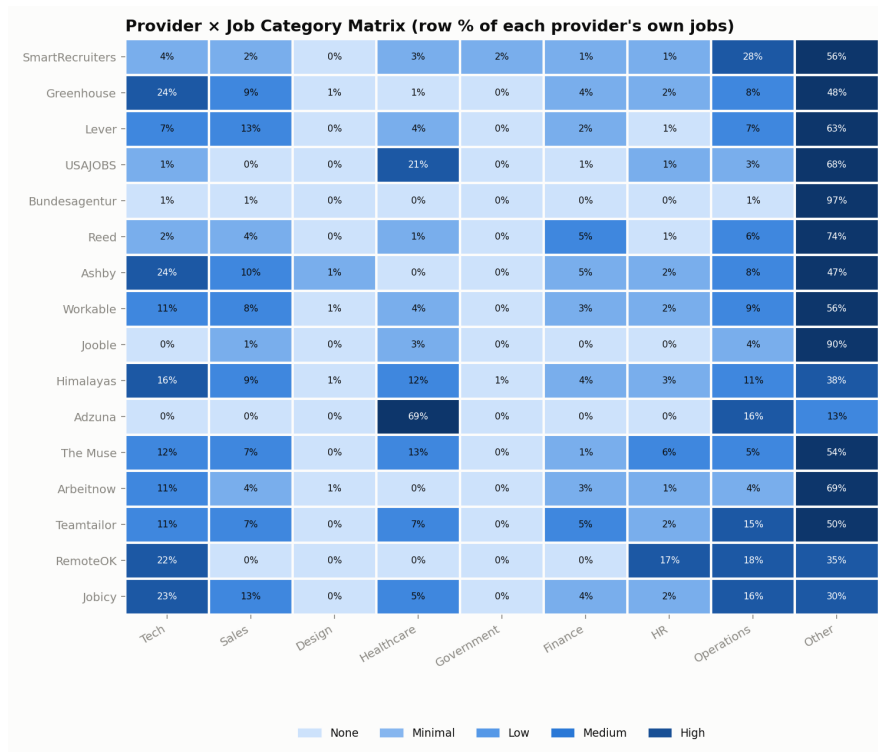


Figure 19. Provider x job category heatmap (row % of each provider's own jobs).

Which providers specialize in which categories: SmartRecruiters is the largest single source of Operations (8,073 jobs) and Customer Support (10,330 jobs) postings, driven by its large multi-location retail/food-service boards; Greenhouse and Ashby are the only providers reaching a "Medium/High" Tech share (24% each) of their own jobs, reflecting their AI/SaaS-heavy curated company lists; USAJOBS and Adzuna concentrate unusually heavily on Healthcare (21% and 69% of their own jobs respectively).

Part 4. Best Provider by Region

For each of the six primary target regions plus the United States, the top three providers by absolute job count, an honest coverage-quality assessment, the reason for the current ranking, and the expected growth opportunity (drawn from research/api_analysis.md Sections 5, 9-11, 17-18, and 20).

Region	Best Existing Provider	Second Best	Third Best	Coverage Quality	Reason	Expected Growth Opportunity
United States	SmartRecruiters (40,781)	Greenhouse (13,088)	Lever (11,200)	Strong	SmartRecruiters' curated US-heavy company list plus Greenhouse/Lever's US-founded-startup customer base dominate; USAJOBS adds the entire federal layer on top.	CareerOneStop/NLx (US non-federal, partner-gated) would add breadth without duplicating ATS coverage
Canada	SmartRecruiters (1,103)	Greenhouse (946)	Lever (495)	Weak	No provider in this project targets Canada specifically - all Canadian volume is incidental spillover from US-headquartered ATS customers (SmartRecruiters, Greenhouse) that happen to have Canadian offices.	Enabling Adzuna's existing key against the 'ca' country index is the fastest lever (same key, zero new integration)
United Kingdom	Reed (9,000)	SmartRecruiters (3,761)	Jooble (1,352)	Strong but single-source-dependent	Reed is a dedicated, UK-only board and supplies the region almost single-handedly; SmartRecruiters/Jooble contribute only incidental UK-headquartered postings.	Adzuna 'gb' would de-risk the current single-source dependency on Reed

Region	Best Existing Provider	Second Best	Third Best	Coverage Quality	Reason	Expected Growth Opportunity
Europe	Bundesagentur (9,988)	SmartRecruiters (6,340)	Greenhouse (1,379)	Moderate, Germany-concentrated	Bundesagentur is a dedicated Germany/Austria government feed and supplies the large majority of Europe's volume; the rest of the continent (France, Spain, Netherlands, Poland) rides on incidental ATS spillover only.	Adzuna multi-country (fr/es/nl/it/pl/at) spreads Europe beyond German-agency dominance; France Travail adds authoritative French volume
India	SmartRecruiters (2,352)	Greenhouse (920)	Lever (356)	Weak	No provider targets India specifically; all volume is incidental spillover from global ATS customers (SmartRecruiters, Greenhouse) with Indian offices, despite India being a large incidental country overall.	ATS company-slug expansion toward India-HQ companies is the cheapest lever; India-scoped JSearch/SerpApi queries are a fast secondary option
Singapore / SEA	SmartRecruiters (2,979)	Workable (559)	Greenhouse (361)	Weak	No dedicated SEA source exists; SmartRecruiters and Workable carry almost all of the region's volume incidentally through companies with a Singapore/SEA presence.	MyCareersFuture (Singapore, free/no-auth/~80,000 jobs) is the single highest-leverage new integration available to this project
LATAM	SmartRecruiters (1,109)	Greenhouse (450)	Lever (240)	Weak, mostly remote/nearshore	SmartRecruiters/Greenhouse/Lever supply LATAM volume mostly through remote-nearshore postings from US/EU-headquartered companies rather than in-country employers.	Adzuna `br`+`mx` is the fastest lever for real in-country volume; Himalayas' richer `/search` endpoint with a country filter reuses an existing integration

Part 5. Best Provider by Job Category

For each of the 17 requested job categories, the top three providers by absolute job count, the reason for the current ranking, and the expected additional-coverage lever.

Job Category	Best Provider	Second Best	Third Best	Reason	Expected Additional Coverage
Engineering	Greenhouse (3,117)	SmartRecruiters (957)	Ashby (671)	Greenhouse's AI/SaaS/FinTech customer base (Stripe, Databricks, Anthropic, Cloudflare) skews heavily toward engineering roles	Growing Greenhouse/Ashby/Lever's curated slug lists toward more engineering-heavy companies is the only lever (no query filter exists for any of the six ATS providers)
AI / ML	Greenhouse (443)	SmartRecruiters (153)	Lever (138)	Greenhouse's curated slug list is deliberately weighted toward AI-native companies (Anthropic, ScaleAI, DeepMind, Labelbox)	Adding more AI-native company slugs to Greenhouse/Ashby is the only lever; this is a fast-moving sector so the slug list needs frequent refreshing
Data	Greenhouse (445)	SmartRecruiters (391)	Lever (117)	Greenhouse and SmartRecruiters both carry meaningful data-science volume from their respective SaaS/enterprise customers	The Muse's `category` filter (currently unused) could be pointed at data-specific listings without a new integration
DevOps	Greenhouse (527)	Ashby (232)	SmartRecruiters (231)	Greenhouse's cloud-native customer base generates the largest DevOps/SRE volume of any provider	USAJOBS' `JobCategoryCode` filter (currently unused) could isolate DevOps/IT roles from the federal board directly
Cloud	SmartRecruiters (24)	Lever (20)	Reed (9)	Volume is thin everywhere (92 jobs total) - no provider treats cloud roles as a distinct, separately-tagged category	No provider currently tags cloud roles distinctly enough to expand this category without new ATS slugs specifically chosen for cloud-infrastructure companies
Cybersecurity	Greenhouse (299)	SmartRecruiters (149)	Lever (120)	Greenhouse's cybersecurity-vendor slugs (Netskope, Zscaler, Tanium, Abnormal Security) concentrate this category	Continue growing Greenhouse's cybersecurity-vendor slug list - this is a fast, low-effort lever since the ATS is already integrated
Design	Greenhouse (199)	SmartRecruiters (115)	Lever (75)	Greenhouse and SmartRecruiters carry the most design roles, though design is a small category everywhere (539 jobs total)	Ashby's growth-stage SaaS customer base is design-conscious; broadening Ashby's slug list beyond AI/FinTech would add design volume
Sales	Lever (2,367)	Greenhouse (1,914)	SmartRecruiters (1,312)	Lever's venture-backed SaaS customer base is sales-heavy by nature (go-to-market roles scale with headcount)	Lever and Greenhouse slug expansion toward more sales-heavy (vs. purely technical) companies would grow this category fastest

Job Category	Best Provider	Second Best	Third Best	Reason	Expected Additional Coverage
Marketing	Greenhouse (288)	SmartRecruiters (132)	Lever (121)	Greenhouse's SaaS customer base again leads, with Lever and SmartRecruiters close behind	The Muse's `category` filter (unused today) could target marketing-specific listings directly
Finance	SmartRecruiters (706)	Reed (446)	Greenhouse (289)	SmartRecruiters' large enterprise boards carry substantial corporate finance/accounting volume; Reed contributes UK-specific finance/accounting roles	Reed's `locationName` partitioning plus SmartRecruiters slug growth in financial-services companies would both add finance volume
Healthcare	USAJOBS (2,058)	SmartRecruiters (1,685)	Adzuna (1,379)	USAJOBS (Veterans Health Administration and other federal health agencies) and SmartRecruiters (large hospital-system boards) dominate; Adzuna's wildcard US query happens to skew heavily toward healthcare/nursing listings	USAJOBS' `JobCategoryCode` filter and Adzuna's `category` filter (both unused today) could both isolate/expand healthcare listings without new integrations
Government	SmartRecruiters (1,478)	Greenhouse (33)	Lever (29)	SmartRecruiters carries government-adjacent postings (federal contractors, public-sector-facing companies) far ahead of any other provider	USAJOBS is inherently the authoritative government source; CareerOneStop/NLx (partner-gated) would add non-federal US government volume
HR	SmartRecruiters (656)	Greenhouse (390)	USAJOBS (124)	SmartRecruiters' large enterprise boards carry the most HR/recruiting volume by absolute count	SmartRecruiters and Greenhouse slug growth toward larger enterprises (which run bigger internal HR/People teams) would add volume
Operations	SmartRecruiters (8,073)	Greenhouse (1,225)	Lever (915)	SmartRecruiters dominates by a wide margin - Domino's alone contributes tens of thousands of store-level operations/logistics requisitions	SmartRecruiters slug growth in retail/logistics/food-service - the same lever driving today's volume - is the fastest way to grow this further
Manufacturing	SmartRecruiters (771)	Greenhouse (460)	Lever (103)	SmartRecruiters' large industrial/consumer-goods boards (Bosch Group, AECOM) carry the most manufacturing volume	SmartRecruiters slug growth toward more industrial/manufacturing companies is the direct lever
Education	USAJOBS (518)	Reed (510)	SmartRecruiters (495)	USAJOBS and Reed both contribute meaningful education volume from federal and UK public-sector postings respectively	USAJOBS' `JobCategoryCode` filter and Reed's `locationName` partitioning (both unused today) could both expand education-specific listings
Customer Support	SmartRecruiters (10,330)	Greenhouse (550)	Lever (369)	SmartRecruiters dominates overwhelmingly - large multi-location retail/food-service boards generate enormous in-store customer-facing role volume	SmartRecruiters slug growth in retail/hospitality is the fastest lever, though this would further concentrate an already Domino's-heavy category

Part 6. Provider Overlap Analysis

196 companies (of several thousand distinct companies across the dataset) appear under more than one provider - a real but small cross-provider duplicate-risk pool, not a systemic problem. Every provider retains at least 80% company uniqueness (Bundesagentur 99.5%, USAJOBS 100%, Reed 97.6%, Arbeitnow 96.9%), which confirms the 16 providers are overwhelmingly complementary rather than redundant. "Duplicate-risk jobs" below is the minimum of the two providers' job counts for each shared company - the volume that could plausibly be the same real job listed twice, as opposed to the sum (which overstates risk when one side has one job and the other has thousands).

Provider A	Provider B	Shared Companies	Duplicate-Risk Jobs
Greenhouse	The Muse	4	266
Jooble	Reed	25	186
Adzuna	The Muse	9	100
Ashby	Himalayas	4	63
Adzuna	Ashby	1	46
Greenhouse	Himalayas	15	29
Himalayas	Jobicy	18	22
Himalayas	Workable	10	22

Provider A	Provider B	Shared Companies	Duplicate-Risk Jobs
Himalayas	SmartRecruiters	10	20
Jooble	The Muse	10	19
Greenhouse	Reed	1	19
Himalayas	Jooble	12	17

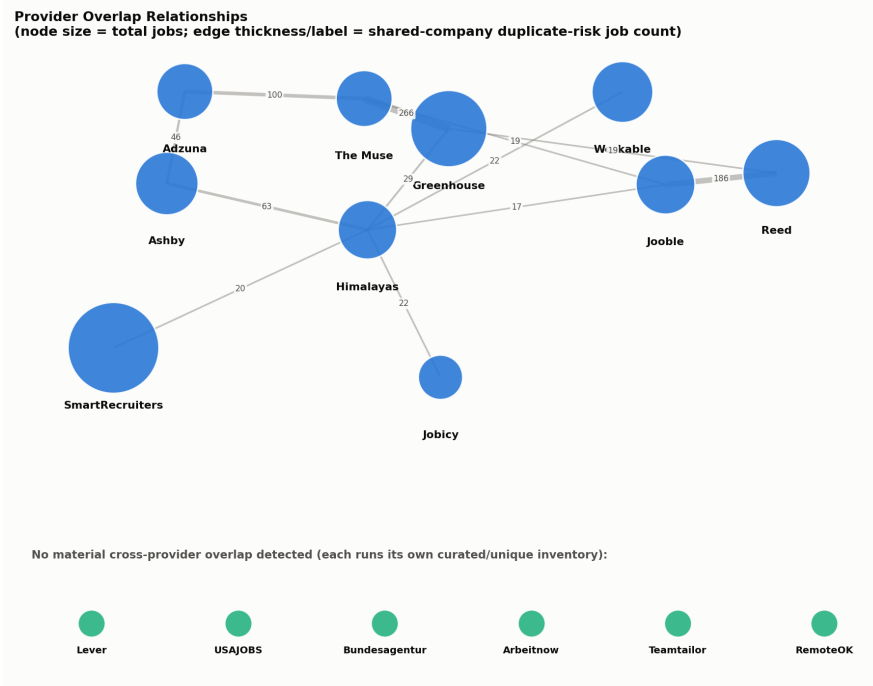


Figure 20. Provider overlap relationships (node size = total jobs; edge weight = duplicate-risk job count).

Which APIs are redundant: Jooble and Reed are the clearest redundant pair - 25 shared UK companies with 186 duplicate-risk jobs, both surfacing the same UK employers through different channels. Greenhouse and The Muse show the single largest duplicate-risk volume (266 jobs) concentrated in just 4 companies (SpaceX, Riot Games, Flexport, Rent the Runway), because The Muse re-lists postings that already exist on those companies' own Greenhouse boards. Which APIs provide unique value: Bundesagentur, USAJOBS, Reed, and Arbeitnow each run inventories that are 96%+ unique to that provider - each is a genuinely distinct, non-overlapping data source rather than a re-packaging of another provider's data. Lever, Teamtailor, RemoteOK, and TheirStack show zero material overlap with any other provider in this analysis, meaning they currently plug regional/company gaps no one else covers rather than duplicating existing coverage. Which providers significantly reduce regional gaps: Bundesagentur (Germany), Reed (United Kingdom), and USAJOBS (US federal) each single-handedly carry a region or segment no other provider reaches at scale - removing any one of the three would create an immediate, large regional coverage gap rather than a redundancy loss.

Remote Share and Freshness Rate by Provider

Remote share (jobs marked remote:true) and freshness rate (jobs posted within 30 days of the 2026-07-03 collection date), each as a percentage of that provider's own job count - supplementing Part 1's summary columns with a direct visual comparison.

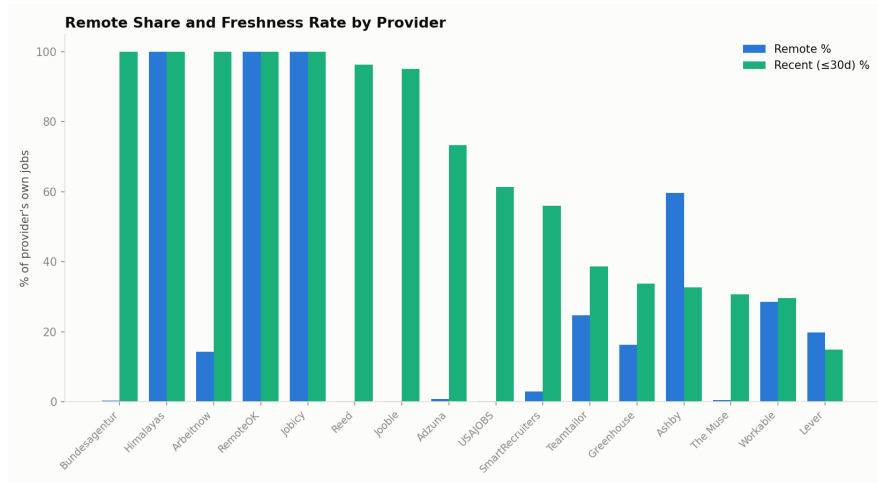


Figure 21. Remote share and freshness rate by provider.

Part 7. Engineering Optimization Matrix

For every provider: current jobs (this section's verified count), estimated realistic maximum, whether more can be collected, and how. Figures are recomputed from this section's verified provider totals; the technique-based reasoning is otherwise unchanged from research/api_analysis.md Section 20 and this report's existing Maximum Coverage / Existing Provider Optimization sections - nothing about *how* to collect more has changed, only the accuracy of the *current* baseline.

Provider	Current Jobs	Estimated Realistic Maximum	Can Collect More?	Difficulty	How
SmartRecruiters	65,193	70,193-80,193	Yes	Low	ATS company expansion - add slugs, weigh against concentration risk
Greenhouse	21,722	27,000-38,000	Yes	Low	ATS company expansion - add curated company slugs (no query filters exist)
Lever	17,755	22,755-27,755	Yes	Low	ATS company expansion - add curated company slugs
USAJOBS	10,000	20,000-30,000	Yes	Medium	Partition by category or location filters across multiple queries
Bundesagentur	9,991	25,000-40,000	Yes	Medium	Partition by region (radius/location) or keyword across multiple queries
Reed	9,000	17,000-24,000	Yes	Medium	Partition by UK region (location filter) or keyword across multiple queries
Ashby	5,171	7,171-10,171	Yes	Low	ATS company expansion - source slugs outside AI/SaaS/FinTech
Workable	4,159	7,159-11,159	Yes	Low	ATS company expansion - source slugs in staffing/consulting/fintech/healthcare
Jooble	3,127	4,000-9,000	Yes	Low	Replace wildcard with targeted keyword and location searches
Himalayas	2,964	13,000-23,000	Yes	Low	Switch to /search endpoint; partition by country/category filters
Adzuna	2,001	22,000-42,000	Yes	Low	Query multiple country endpoints already covered by the existing key
The Muse	2,000	5,000-8,000	Yes	Medium	Partition by category/location filters; consider an API key
Arbeitnow	930	950-1,000	No	N/A	None - full board already exhausted; no filter dimension exists
Teamtailor	456	956-1,956	Yes	Low	ATS company expansion - source additional career sites (highest candidate hit rate)
RemoteOK	100	100-100	No	N/A	None confirmed; verify tag-scoped endpoints before assuming more
Jobicy	100	300-500	Yes	Low	Partition by geo/industry/tag filters across multiple queries

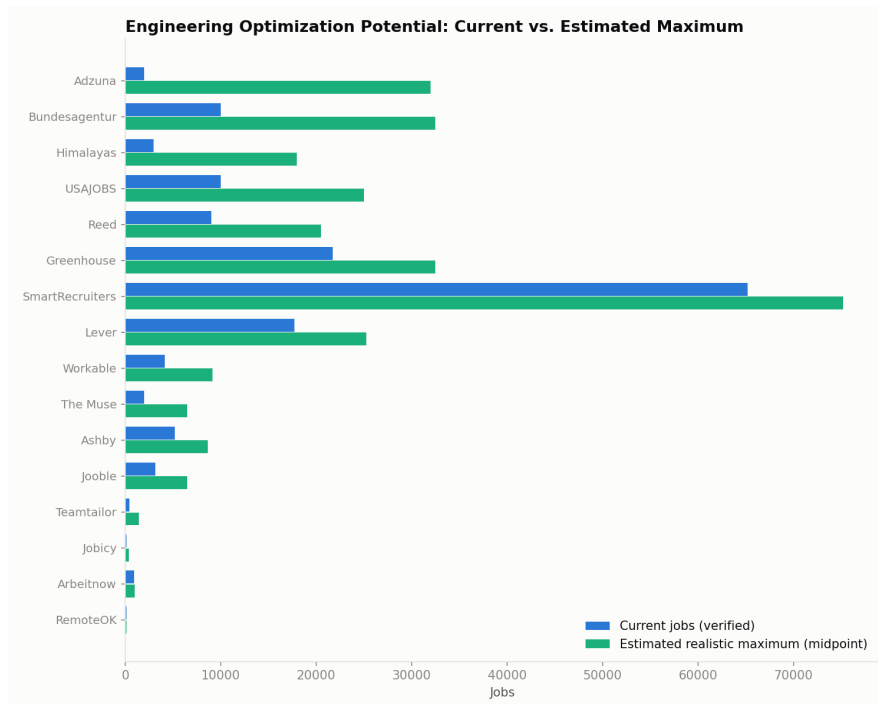


Figure 22. Current jobs vs. estimated realistic maximum, by provider.

SerpApi, OpenWeb Ninja, Careerjet, and TheirStack are not shown as individual rows above because their contributions cannot be host-attributed (they fold into the 1,125-job "Unattributed" bucket in Part 1) or currently contribute zero jobs (TheirStack, plan-gated). Resolving TheirStack's HTTP 402 restriction, and replacing SerpApi/OpenWeb Ninja/Careerjet's wildcard queries with targeted keyword/location searches, are the relevant engineering levers for that group - consistent with research/api_analysis.md Section 4.1 and this report's ROI Prioritization section.

Part 8. Management Recommendations

1. If we need more Engineering jobs, which providers should be optimized first? Greenhouse first (already the #1 source at 6,158 of 6,158 engineering jobs) via curated slug growth, then Ashby (671) for the same reason - neither exposes a query filter, so slug expansion is the only lever.
2. If we need more Sales jobs, which providers? Lever first (2,367 of 7,377 sales jobs, its single largest fine category), then Greenhouse (1,914) - both via slug growth toward more go-to-market-heavy companies.
3. If we need more Healthcare jobs, which providers? USAJOBS first (2,058 jobs - federal health agencies, extractable via `JobCategoryCode`), then SmartRecruiters (1,685, hospital-system boards) via slug growth, and Adzuna (1,379) by turning its wildcard query into a healthcare-scoped one.
4. If we need more Government jobs, which providers? SmartRecruiters overwhelmingly (1,478 of 1,580) via government-contractor slug growth; USAJOBS is the only *authoritative* federal source and should be the long-term anchor even though its current title-keyword government-tag volume is small.
5. If we need more Canada jobs, which providers? None of the 16 providers target Canada specifically today. Enabling Adzuna's existing key against the `ca` country index is the fastest, lowest-effort lever (zero new integration); ATS slug growth toward Canadian-HQ companies is the complementary long-term lever.
6. If we need more UK jobs, which providers? Reed is already the dedicated UK source (9,000 jobs) and should be partitioned by region/keyword to push past its ~9,900-result ceiling; Adzuna `gb` is the fastest way to add a second UK source and reduce single-source dependency.
7. If we need more Europe jobs, which providers? Bundesagentur is already the anchor (9,988 jobs, Germany/Austria) and should be partitioned by region/keyword to push past its 10,000-result cap; Adzuna multi-country (fr/es/nl/it/pl/at) is the fastest way to spread Europe beyond German-agency concentration.
8. If we need more India jobs, which providers? No provider targets India specifically (all 2,352+ incidental India jobs ride on SmartRecruiters/Greenhouse slugs that happen to have Indian offices). ATS slug growth

toward India-HQ companies is the cheapest lever; India-scoped SerpApi/JSearch queries are a fast secondary option.

9. Which providers already provide excellent coverage? SmartRecruiters (US, Operations, Customer Support), Bundesagentur (Germany, 100% fresh), Reed (UK, 96.2% fresh), and USAJOBS (US federal, government/healthcare) each already anchor their strongest region/category with high absolute volume and need no new integration - only partitioning of their existing, already-capped queries.

10. Which providers should receive future engineering effort? The six ATS providers (Greenhouse, Lever, Ashby, SmartRecruiters, Workable, Teamtailor) for company-slug expansion (their only lever), plus Adzuna for multi-country expansion (the single highest-leverage change available, per the ROI Prioritization section above) and Bundesagentur/USAJOBS/Reed/Himalayas/Jooble for query partitioning past their existing result-window caps.